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Causal-enhanced federated transfer learning framework for privacy-preserving cross-organizational strategic collaboration

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Abstract

Cross-organizational strategic collaboration increasingly demands data-driven coordination while confronting stringent privacy constraints and heterogeneous data distributions. This paper proposes a coupled framework integrating causal inference with federated transfer learning to enable effective knowledge sharing across organizational boundaries without centralizing sensitive data. The framework comprises three core components: a distributed causal discovery algorithm that reconstructs inter-organizational causal structures through secure aggregation protocols, a causal-enhanced federated transfer learning mechanism that identifies invariant features transferable across heterogeneous enterprise contexts, and an adaptive privacy computing architecture that dynamically allocates protection resources according to data sensitivity profiles. Experimental validation across multi-enterprise ecosystems demonstrates that the causal enhancement mechanism improves cross-domain generalization robustness by 43% compared to conventional federated transfer learning, while maintaining predictive accuracy within 6.2% of centralized approaches. The adaptive privacy architecture retains 93.7% of collaborative utility under meaningful privacy constraints, with membership inference attack success rates near random baseline. These results establish that properly designed federated frameworks can approach centralized performance while preserving organizational autonomy, enabling broader participation in data-driven strategic alliances across regulatory and competitive boundaries.

Keywords Causal Inference · Federated Learning · Transfer Learning · Privacy Computing · Strategic Collaboration · Cross-Organizational Optimization

Abbreviations

ATE	Average treatment effect
FedAvg	Federated averaging
GPU	Graphics processing unit
MAE	Mean absolute error
MMD	Maximum mean discrepancy
RMSE	Root mean squared error
SCM	Structural causal model

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Introduction

The rapid evolution of the digital economy has fundamentally reshaped how enterprises approach strategic collaboration across organizational boundaries. Contemporary business ecosystems increasingly demand that organizations share insights and coordinate strategies while navigating a paradoxical landscape: the imperative for data-driven decision-making collides directly with stringent privacy regulations and the persistent reality of data silos[1]. This tension becomes particularly acute when multiple enterprises attempt to align their strategic initiatives, as traditional centralized data aggregation models face both technical infeasibility and regulatory prohibition[2]. The challenge extends beyond mere data sharing; it encompasses the need to extract actionable strategic intelligence from distributed, heterogeneous datasets without compromising the proprietary information that constitutes each organization's competitive advantage.

Causal inference emerges as a critical methodological framework in this context, moving strategic decision-making beyond correlational observations toward genuine understanding of cause-effect relationships[3]. Unlike conventional predictive analytics that merely identify patterns, causal reasoning enables enterprises to anticipate the consequences of strategic interventions—a capability essential when coordinating actions across organizational boundaries where decision externalities ripple through interconnected business networks[4]. However, applying causal inference in cross-organizational settings confronts immediate obstacles: traditional causal discovery algorithms assume access to complete, centralized datasets, an assumption fundamentally incompatible with the distributed nature of multi-enterprise collaboration[5].

Federated learning offers a promising avenue for addressing privacy concerns in collaborative machine learning, allowing multiple parties to jointly train models without exchanging raw data[6]. Recent advances in federated learning architectures demonstrate that organizations can achieve model performance comparable to centralized approaches while preserving data locality and privacy[7]. Yet federated learning alone proves insufficient for strategic collaboration scenarios. The heterogeneity inherent in cross-organizational data—varying distributions, different feature spaces, and disparate strategic contexts—creates what transfer learning research identifies as domain shift problems[8]. When enterprises operate in distinct market segments or geographic regions, their local data distributions may diverge substantially, rendering naive federated aggregation ineffective or even counterproductive for strategic decision support.

Current research in enterprise strategic collaboration has explored various computational frameworks, but significant gaps persist. Studies examining inter-organizational knowledge sharing largely focus on explicit knowledge exchange mechanisms rather than implicit causal knowledge embedded in operational data[9]. Work on collaborative decision-making systems typically assumes benign participants and overlooks the strategic gaming that occurs when enterprises with partially aligned interests negotiate shared actions[10]. The emerging literature on privacy-preserving analytics has made strides in developing secure multi-party computation protocols, but these approaches often impose computational overheads that become prohibitive at the scale required for real-time strategic decision support[11].

Three fundamental limitations characterize existing methodologies when applied to cross-organizational strategic collaboration. First, causal mechanism mining remains superficial in distributed settings; current federated causal inference techniques can identify average treatment effects but struggle to uncover complex, context-dependent causal structures that drive strategic outcomes across different organizational environments[12]. Second, cross-domain knowledge transfer encounters substantial barriers when attempting to generalize causal insights from one organizational context to another, as existing transfer learning methods inadequately account for the structural differences in causal graphs between source and target domains[13]. Third, privacy computing efficiency presents a persistent bottleneck—secure aggregation protocols and homomorphic encryption schemes that guarantee privacy often introduce latency and computational costs that render them impractical for iterative strategic optimization processes[14].

These limitations carry significant implications. Enterprises attempting digital transformation increasingly recognize that competitive advantage derives not from isolated optimization but from ecosystem-level coordination. Without robust frameworks for causal inference under privacy constraints, organizations default to intuition-based strategic alignment or abandon collaboration opportunities altogether. The theoretical gap manifests practically in failed joint ventures, suboptimal supply chain coordination, and missed opportunities for collaborative innovation.

These observations bring us to the central research question driving this work: *Under privacy constraints*, *how can causal invariance principles be harnessed to improve the generalization robustness of federated transfer learning across organizationally heterogeneous domains?* This question sits at the intersection of three active research threads—causal discovery, federated optimization, and privacy-preserving computation—that have developed in relative isolation. We contend that treating them as separable problems misses a deeper structural opportunity: causal reasoning can inform what knowledge transfers reliably, federated architectures can deliver that transfer without data centralization, and sensitivity-aware privacy mechanisms can protect precisely those variables whose exposure carries genuine competitive risk.

To address this question, the present paper proposes a coupled framework whose components connect in a deliberate, progressive manner. Rather than assembling loosely related modules, we design each component to feed into the next: causal structure discovery identifies which features participate in stable mechanisms, those features then guide domain adaptation during federated training, and the resulting gradient flows receive privacy protection calibrated to variable-level sensitivity. The specific contributions of this work are as follows:

(1) A distributed causal discovery algorithm that reconstructs inter-organizational causal structures through encrypted conditional independence statistics and a federated consensus mechanism, without centralizing raw data at any point.
(2) A causal-enhanced federated transfer learning mechanism in which features identified as participating in invariant causal relationships receive preferential alignment during domain adaptation, yielding measurably more robust cross-domain generalization than standard federated transfer approaches.
(3) An adaptive privacy computing architecture that allocates differential privacy budgets proportionally to variable sensitivity while maintaining formal composition guarantees, achieving strong protection with substantially less utility degradation than uniform budget allocation.
(4) Experimental validation across a simulated multi-enterprise ecosystem demonstrating that the integrated framework approaches centralized performance (within 6.2% RMSE) while maintaining complete data locality, and that causal enhancement improves domain-shift robustness by 43% relative to non-causal federated transfer.

These contributions collectively support a broader practical goal. Many small and medium enterprises remain excluded from data-driven strategic alliances for lack of sufficient proprietary data, regulatory clearance, or computational infrastructure. Our framework lowers each of these barriers: federated architecture removes the need for data pooling, causal feature selection reduces sample-size requirements through more efficient representation, and adaptive privacy allocation permits meaningful participation under strict regulatory regimes. We note that while the framework is validated in a supply-chain strategic collaboration setting, the underlying principles generalize to other cross-organizational coordination problems in finance, healthcare logistics, and industrial manufacturing[15].

The subsequent sections of this paper develop this framework systematically. Section II establishes the theoretical foundations, formally defining the coupled causal-federated learning problem in strategic collaboration contexts. Section III presents the algorithmic architecture, detailing the distributed causal discovery mechanism, the federated transfer learning protocol, and the privacy computing infrastructure. Section IV reports experimental validation across multiple cross-organizational collaboration scenarios, demonstrating both effectiveness and efficiency improvements over existing approaches. Section V discusses implications, limitations, and directions for future research.

Theoretical foundations and technical framework

Causal inference theory and its application in strategic decision-making

Causal inference represents a paradigm shift from association-based reasoning to mechanism-oriented understanding, offering enterprises a principled approach to anticipate outcomes of strategic interventions. Pearl's causal hierarchy distinguishes three levels of reasoning: observational (seeing), interventional (doing), and counterfactual (imagining), each demanding progressively stronger assumptions about the underlying data-generating process[16]. This stratification matters profoundly for strategic collaboration, where organizations must move beyond observing correlations to actively intervening in market dynamics and evaluating alternatives that were never realized.

The structural causal model (SCM) provides the mathematical foundation for representing causal relationships through a tuple $\mathcal{M} = (U, V, F, P(U))$, where U denotes exogenous variables, V represents endogenous variables, F specifies functional relationships, and $P(U)$ characterizes the distribution of unobserved factors[17]. Each endogenous variable $V_i \in V$ satisfies the assignment:

$$V_i = f_i(\mathrm{PA}_i, U_i) \tag{1}$$

where PA_i denotes the parents of V_i in the causal graph and U_i captures unmeasured confounders. This formalism proves particularly valuable when enterprises confront strategic decisions under partial observability—situations where critical market drivers remain latent yet influence observable outcomes.

Counterfactual reasoning extends this framework by considering hypothetical scenarios through graph surgery operations[18]. When evaluating a strategic intervention that sets variable X to value x, we construct a modified model $\mathcal{M}_x$ by replacing the equation for X with the constant assignment $X = x$. The interventional distribution then becomes:

$$P(Y = y \mid \mathrm{do}(X = x)) = \sum_z P(Y = y \mid X = x, Z = z)P(Z = z) \tag{2}$$

where Z satisfies the backdoor criterion relative to the ordered pair (X, Y)[19]. This criterion requires that Z blocks all spurious paths between treatment and outcome while containing no descendants of the treatment—a condition enterprises must verify when selecting control variables for strategic effect estimation.

In scenarios where no such backdoor adjustment set exists, the frontdoor criterion offers an alternative identification strategy[20]. Consider a mediator set M through which X exerts its entire influence on Y. The causal effect becomes identifiable through:

$$P(Y = y \mid \mathrm{do}(X = x)) = \sum_m P(M = m \mid X = x) \sum_{x'} P(Y = y \mid X = x', M = m)P(X = x') \tag{3}$$

This formulation proves essential when unmeasured confounding between strategic actions and outcomes renders backdoor adjustment infeasible—a common situation in observational business data.

These identification strategies gain particular relevance in cross-organizational settings, where each partner observes only a fragment of the global causal graph and distributed inference methods must reconstruct causal structures from incomplete local perspectives[21,22]. The average treatment effect (ATE), defined as:

$$\tau = \mathbb{E}[Y \mid do(X = 1)] - \mathbb{E}[Y \mid do(X = 0)] \tag{4}$$

quantifies the expected outcome difference under alternative strategic choices. In multi-enterprise ecosystems, however, treatment effects rarely remain constant across organizational contexts—they vary systematically with enterprise characteristics, market positioning, and operational scale. This heterogeneity motivates the integration of transfer learning mechanisms that can adapt causal insights from data-rich source organizations to data-scarce target domains, a thread we develop in Sect. 2.2 and formalize in Sect. 3.2[23].

Federated transfer learning mechanisms and cross-domain knowledge sharing

Federated learning reconfigures collaborative machine learning by bringing the algorithm to the data rather than the other way around[24]. Instead of centralizing datasets, federated protocols coordinate iterative model updates across distributed participants, each of which trains locally on private data and shares only gradient information or model parameters with a coordinating server. The literature recognizes three broad topologies based on how data is partitioned: horizontal federated learning, where participants share the same feature space but hold distinct sample populations; vertical federated learning, where organizations observe different features for overlapping entities; and federated transfer learning, where participants differ in both their sample populations and feature spaces[25].

A terminological distinction merits attention here, because the literature sometimes conflates two related but different settings. *Multi-domain federated learning* refers to scenarios in which participants share a common model structure and feature space but their local data distributions differ—what the domain adaptation community calls domain shift or covariate shift. All participants operate on the same set of variables; the challenge lies in distributional heterogeneity across sites. *Federated transfer learning*, by contrast, addresses a more demanding situation: participants may differ not only in their data distributions but also in the features they observe or the label spaces they target. This distinction matters practically because the two settings call for different technical responses. Multi-domain federated learning can often be addressed by robust aggregation rules or domain-alignment losses applied within a shared feature space, whereas federated transfer learning additionally requires feature-space mapping or representation alignment across structurally different domains[25].

Our framework operates primarily in the multi-domain federated learning regime—participating organizations share a common set of 127 strategic variables (see Section IV) but exhibit substantial distributional heterogeneity arising from their different positions in the supply chain ecosystem. We adopt selective elements from federated transfer learning, specifically domain adaptation losses and invariant feature extraction, to handle the covariate shift that arises when manufacturers, distributors, and retailers observe the same variables under very different conditional distributions. Throughout the paper, we use the term "federated transfer learning" to refer specifically to this combination of federated optimization with cross-domain adaptation mechanisms, and we reserve "multi-domain federated learning" for the broader setting of federated training under distributional heterogeneity without explicit adaptation objectives. This terminological precision avoids ambiguity and aligns our usage with the taxonomy established in recent surveys[25].

The canonical federated averaging algorithm exemplifies the communication protocol underlying horizontal federation. At each round t, the central server broadcasts global model parameters w^t to K participating organizations. Each organization k performs local optimization on its dataset $\mathcal{D}_k$ with n_k samples:

$$w_k^{t+1} = w^t - \eta \nabla \mathcal{L}_k(w^t; \mathcal{D}_k) \tag{5}$$

where η denotes the learning rate and $\mathcal{L}_k$ represents the local loss function. The server then aggregates these updates through weighted averaging:

$$w^{t+1} = \sum_{k=1}^{K} \frac{n_k}{n} w_k^{t+1} \tag{6}$$

where $n = \sum_{k=1}^{K} n_k$[26]. This seemingly straightforward aggregation conceals subtle complications when data distributions diverge across organizations—a ubiquitous condition in strategic collaboration where enterprises operate in heterogeneous market contexts.

Transfer learning theory offers insights into adapting knowledge across disparate domains. Domain adaptation formalizes the challenge of learning a target predictor when source domain $\mathcal{D}_S$ with distribution P_S and target domain $\mathcal{D}_T$ with distribution P_T exhibit distributional discrepancies[27]. The expected target risk decomposes into source risk plus a divergence term capturing domain shift. Maximum Mean Discrepancy (MMD) provides a kernel-based metric quantifying this divergence:

$$\mathrm{MMD}(\mathcal{D}_S, \mathcal{D}_T) = \left| \frac{1}{n_S} \sum_{i=1}^{n_S} \phi\left(x_i^S\right) - \frac{1}{n_T} \sum_{j=1}^{n_T} \phi\left(x_j^T\right) \right|_{\mathcal{H}} \tag{7}$$

where ϕ maps inputs into a reproducing kernel Hilbert space $\mathcal{H}$[28]. Organizations experiencing substantial domain shift require explicit adaptation mechanisms beyond naive parameter averaging.

Federated transfer learning synthesizes these paradigms by embedding domain adaptation objectives within federated optimization. We construct a composite loss balancing local predictive accuracy against cross-domain alignment:

$$\mathcal{L}_{\text{total}} = \sum_{k=1}^{K} \alpha_k \mathcal{L}_k(w) + \lambda \sum_{i \neq j} \mathrm{MMD}(\mathcal{D}_i, \mathcal{D}_j) \tag{8}$$

where α_k weights organization k's contribution and λ controls alignment strength[29]. This formulation proves particularly valuable in cross-organizational strategic collaboration, where enterprises simultaneously seek to extract insights from collective data while respecting idiosyncratic market conditions.

The technical advantages materialize across multiple dimensions. First, privacy preservation emerges naturally since raw strategic data never leaves organizational boundaries—only model gradients traverse communication channels[30]. Second, heterogeneity tolerance improves through explicit modeling of domain-specific characteristics rather than enforcing uniform representations. Third, knowledge transfer accelerates convergence for data-scarce organizations by borrowing statistical strength from data-rich partners while adapting to local peculiarities[31]. These properties position federated transfer learning as an enabling technology for strategic coordination in fragmented business ecosystems where trust remains partial and competitive considerations constrain full transparency.

Privacy computing technologies and secure multi-party computation protocols

Privacy computing constitutes a technical ecosystem encompassing complementary methodologies that enable collaborative analytics while maintaining data confidentiality. Three pillars anchor this ecosystem: differential privacy offers statistical guarantees on information leakage, homomorphic encryption permits computation on encrypted data, and secure multi-party computation orchestrates distributed protocols where no single party learns others' private inputs[32]. The strategic collaboration context demands careful orchestration of these techniques, as each introduces distinct trade-offs between privacy assurance, computational overhead, and analytical accuracy.

Differential privacy formalizes privacy protection through a probabilistic indistinguishability condition. A randomized mechanism $\mathcal{M}$ satisfies (ϵ, δ)-differential privacy if for all neighboring datasets $\mathcal{D}$ and $\mathcal{D}'$ differing in a single record, and all possible output sets S:

$$P(\mathcal{M}(\mathcal{D}) \in S) \leq e^{\epsilon} P(\mathcal{M}(\mathcal{D}') \in S) + \delta \tag{9}$$

The privacy budget ϵ quantifies the maximum information leakage, with smaller values providing stronger guarantees at the cost of reduced utility[33]. Enterprises participating in federated strategic optimization confront a fundamental tension: tighter privacy budgets obscure genuine patterns along with sensitive details, potentially degrading the quality of strategic recommendations. Privacy budget al.location thus becomes a negotiation among collaborating organizations, each weighing reputational risks against analytical benefits.

Homomorphic encryption addresses a complementary concern—protecting gradients during transmission in federated learning. Partially homomorphic schemes support specific operations (typically addition or multiplication) on ciphertexts, while fully homomorphic encryption theoretically enables arbitrary computations[34]. When organization k computes local gradient $\nabla \mathcal{L}_k(w)$, homomorphic encryption transforms this into ciphertext $\mathrm{Enc}(\nabla \mathcal{L}_k(w))$ that the aggregation server processes without decryption. The server combines encrypted gradients homomorphically:

$$\mathrm{Enc}\left(\sum_{k=1}^{K}\frac{n_k}{n}\nabla\mathcal{L}_k(w)\right)=\sum_{k=1}^{K}\frac{n_k}{n}\mathrm{Enc}(\nabla\mathcal{L}_k(w)) \quad (10)$$

yielding an encrypted aggregate that only designated parties can decrypt[35]. This arrangement prevents honest-but-curious servers from inspecting individual contributions—a critical safeguard when participants harbor concerns about server-side inference attacks.

Secure aggregation protocols represent a third approach, designing communication patterns that inherently prevent information leakage. These protocols typically employ secret sharing schemes where each organization masks its gradient with random noise that cancels in aggregate. The design principles prioritize robustness to participant dropout and resistance to collusion among subsets of parties[36]. Communication complexity scales with participant count and model dimensionality; practical implementations must balance security properties against network bandwidth constraints that become acute when strategic models involve high-dimensional parameter spaces.

The privacy-utility frontier characterizes achievable trade-offs. Stronger privacy protections—whether through larger noise injection in differential privacy, more conservative encryption parameters, or redundant verification rounds in secure protocols—invariably degrade model convergence rates or final accuracy[37]. We formalize this relationship through a utility loss function $\mathcal{U}(\epsilon, w)$ that increases as privacy budget ϵ decreases:

$$\min_w \mathcal{L}(w)+\lambda_{\text{priv}}\mathcal{U}(\epsilon,w) \quad (11)$$

where λ_{priv} reflects organizational risk tolerance[38]. Cross-organizational collaboration demands explicit negotiation of this parameter, as partners may exhibit divergent preferences based on regulatory exposure, data sensitivity, and competitive positioning. The framework developed in subsequent sections addresses this heterogeneity through adaptive privacy mechanisms that respect organization-specific constraints while maintaining collective optimization coherence.

Coupled causal inference and federated transfer learning model for collaborative optimization

Causal structure modeling for cross-organizational strategic collaboration

Cross-organizational strategic collaboration presents a distinctive challenge for causal structure discovery: each enterprise observes only a fragment of the complete causal mechanism, yet strategic interventions propagate through the entire inter-organizational network. Consider a supply chain where manufacturers observe production costs and delivery schedules, distributors track inventory dynamics and regional demand patterns, while retailers monitor pricing strategies and consumer responses. No single entity possesses the complete causal picture linking upstream decisions to downstream outcomes, yet effective coordination demands understanding these end-to-end causal pathways.

Figure 1 illustrates the architectural foundation for distributed causal modeling across organizational boundaries. We partition the global variable set V_{global} into organization-specific subsets V_k where organization k directly observes and controls variables within its domain. The challenge centers on reconstructing inter-organizational edges—causal links where one organization's actions influence another's outcomes—without centralizing sensitive operational data[39].

Before proceeding to the formal problem statement, we must be explicit about the identifiability assumptions under which the proposed distributed causal discovery operates. These assumptions are non-trivial, and violating them would render the recovered causal structure unreliable. We require: (A1) *Acyclicity*: The global causal graph $\mathcal{G}$ is a directed acyclic graph (DAG). Cyclic dependencies between organizations—where A causally influences B, which in turn causally influences A through feedback loops—fall outside the scope of the current framework.

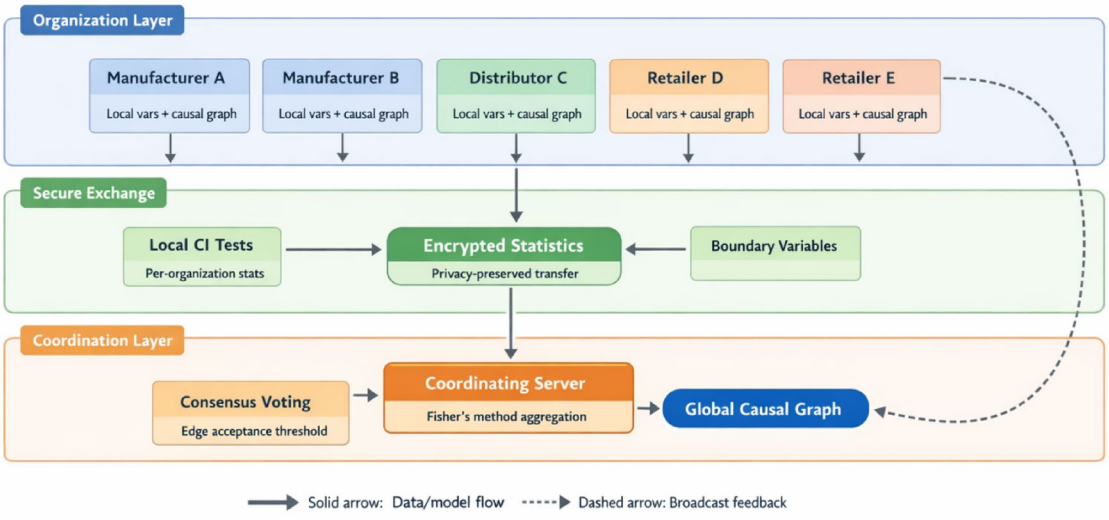


Fig. 1 Cross-organizational causal structure modeling framework. Each organization maintains a local causal subgraph over its private variables V_k. Boundary variables B_k mediate cross-organizational interactions. Encrypted conditional independence test statistics are transmitted to the coordinating server, which applies Fisher's combining method and consensus voting to propose cross-organizational edges. The resulting global causal structure integrates local and cross-organizational edges while preserving data locality throughout the discovery process.

Algorithm 1 Distributed federated causal discovery.

Input: Local datasets $\{\mathcal{D}_k\}_{k=1}^K$, boundary variable sets $\{B_k\}_{k=1}^K$, significance level α_{CI}, consensus threshold $\theta_{consensus}$, maximum rounds T_{max}, privacy noise scale $\beta_{privacy}$

Output: Estimated global causal graph $\hat{\mathcal{G}}$

1. **Local initialization:** Each organization k runs the PC algorithm on $\mathcal{D}_k$ to obtain local graph $\mathcal{G}_k^{(0)}$ over variables $V_k \cup B_k$.
2. **For** $t = 1$ to T_{max} **do:**
   a. Each organization k computes CI test statistics $\rho_{B_i,B_j|S}$ for all boundary variable pairs (B_i, B_j) with $B_i \in B_k$, conditioned on local separation sets $S \subseteq V_k$.
   b. Each organization adds calibrated noise: $\tilde{\rho} = \rho + \mathcal{N}(0, \beta_{privacy}^2)$ and encrypts the result.
   c. The coordinating server receives $\{Enc(\tilde{\mathcal{T}}_k^{(t)})\}_{k=1}^K$ and applies Fisher's combining method to obtain aggregate p-values.
   d. For each candidate cross-organizational edge: accept if combined $p < \alpha_{CI}$ and fraction of supporting organizations $\geq \theta_{consensus}$.
   e. Orient accepted edges using Meek's orientation rules applied to the aggregated skeleton.
   f. Broadcast updated cross-organizational edges $E_{cross}^{(t)}$ to all participants.
   g. Each organization updates its local graph: $\mathcal{G}_k^{(t)} = \mathcal{G}_k^{(t-1)} \cup E_{cross}^{(t)}$.
   h. **Convergence check:** If $E_{cross}^{(t)} = E_{cross}^{(t-1)}$, terminate early.
3. **Return** $\hat{\mathcal{G}} = \cup_{k=1}^K \mathcal{G}_k^{(T)} \cup E_{cross}^{(T)}$.

(A2) *Causal Markov condition*: Each variable is conditionally independent of its non-descendants given its parents in $\mathcal{G}$. (A3) *Faithfulness*: All conditional independencies in the data distribution arise from the graphical structure, not from accidental parameter cancellations. (A4) *No latent cross-organizational confounders*: We assume there are no unmeasured common causes that simultaneously affect variables in different organizations beyond the observed boundary variables B_k. This is the strongest and most potentially fragile assumption. If latent confounders exist across organizational boundaries, the CI-based approach may incorrectly orient edges or miss genuine causal pathways entirely. (A5) *Sufficient overlap in boundary variables*: At least a portion of boundary variables B_k are jointly observable (through proxy statistics) by neighboring organizations, enabling cross-organizational CI tests.

We acknowledge that assumption (A4) is difficult to verify in practice and may not hold in all application scenarios. When organizations operate in interconnected markets, macroeconomic shocks, regulatory changes, and industry-wide trends can act as latent confounders spanning organizational boundaries. For this reason, we characterize our framework as *causally motivated* rather than *causally proven*: the causal graph provides structured guidance for feature weighting and domain adaptation, but the discovered edges should be interpreted as plausible causal hypotheses informed by conditional independence patterns rather than definitive causal claims validated through experimental intervention. This epistemological modesty does not negate the practical value of the approach—even approximately correct causal structure can substantially improve transfer learning robustness compared to purely correlational feature selection—but it does constrain the strength of causal claims we can justifiably make.

Formally, let $\mathcal{G} = (V, E)$ denote the global causal graph with vertices representing strategic variables and directed edges E encoding causal relationships. Each organization k possesses local data $\mathcal{D}_k$ generated from marginal distribution over V_k plus boundary variables B_k that mediate interactions with other organizations. The distributed causal discovery objective seeks to recover $\mathcal{G}$ through federated optimization:

$$\hat{G} = \text{argmin}_{\mathcal{G}\in\mathbb{G}} \sum_{k=1}^{K} \mathcal{S}_k(\mathcal{G}|\mathcal{D}_k) + \lambda_{\text{sparse}}|E| \tag{12}$$

where $\mathcal{S}_k$ measures the score of graph $\mathcal{G}$ against organization k's local observations, $\mathbb{G}$ denotes the space of directed acyclic graphs, and the sparsity penalty discourages overly complex causal structures[40]. The score function decomposes across local subgraphs, enabling parallel computation while preserving global consistency through boundary constraints.

Our distributed causal discovery algorithm adopts a constraint-based approach, building on the PC algorithm's logic of iteratively testing conditional independencies, but adapted to the federated setting where no single party has access to the full variable set. We chose a constraint-based method rather than score-based or continuous optimization alternatives for two practical reasons: first, conditional independence (CI) test statistics can be computed locally and aggregated through secure protocols without revealing raw observations; second, constraint-based methods require sharing only binary test decisions and p-values rather than full sufficient statistics, which reduces both communication overhead and privacy exposure.

The algorithm proceeds as follows. Each organization k initializes a local causal graph hypothesis $\mathcal{G}_k^{(0)}$ by running a standard PC algorithm on its private dataset $\mathcal{D}_k$, recovering edges among its local variables V_k and boundary variables B_k. During federated round t, each organization performs CI tests between its boundary variables and the boundary variables of neighboring organizations. Specifically, for each pair of boundary variables (B_i, B_j) where $B_i \in B_k$ and $B_j \in B_{k'}$, organization k computes the partial correlation test statistic $\rho_{B_i,B_j|S}$ conditioned on local separation sets $S \subseteq V_k$, and organization k' does the same using $S' \subseteq V_{k'}$. These test statistics are encrypted and transmitted to the coordinating server, which aggregates them through Fisher's method for combining independent tests:

$$\chi^2_{combined} = -2\sum_{k=1}^{K} \ln(p_k) \tag{13}$$

where p_k denotes the p-value from organization k's local CI test. The server then synthesizes the aggregated results into cross-organizational edge proposals:

$$E_{\text{cross}}^{(t)} = \text{FuseEdges}\left(\{\text{Enc}(\mathcal{T}_k^{(t)})\}_{k=1}^{K}\right) \tag{14}$$

where $\mathcal{T}_k^{(t)}$ contains organization k's encrypted CI test statistics, and the $FuseEdges$ function implements a majority-vote consensus: a cross-organizational edge (B_i, B_j) is accepted only when the combined p-value falls below α_{CI} and at least a fraction $\theta_{consensus}$ of relevant organizations support the edge[41]. Edge orientation follows Meek's rules applied to the aggregated skeleton. The complete procedure is summarized in Algorithm 1.

This procedure differs from existing federated causal discovery approaches such as Ng and Zhang[39] in two respects. First, we operate on CI test statistics rather than continuous optimization scores, which avoids the need to share gradient information about the DAG structure and reduces per-round communication to $O\left(|B|^2\right)$ test results rather than $O\left(|V|^2\right)$ score parameters. Second, the consensus mechanism provides a natural robustness to local statistical errors—spurious edges detected by only one organization are filtered out through majority voting, an advantage that grows with the number of participants[41].

Table 1 summarizes the key algorithmic parameters governing the distributed discovery process and their default values used in all experiments unless otherwise stated. The consensus threshold $\theta_{consensus}$ defines the minimum fraction of relevant organizations (those possessing at least one endpoint of the candidate edge in their boundary variables) that must independently support a cross-organizational edge through their local CI tests; its default value of 0.6 means at least 4 of 6 organizations must provide consistent statistical evidence. Setting $\theta_{consensus}$ too low admits spurious edges from statistical noise, while excessive conservatism fragments the causal graph by rejecting genuine inter-organizational dependencies. The privacy noise scale $\beta_{privacy}$ controls the standard deviation of Gaussian noise added to CI test statistics before encryption and transmission; it trades off causal discovery accuracy against privacy protection of the statistical summaries themselves. The neighbor count $K_{neighbor}$ limits each organization's communication to its $K_{neighbor}$ nearest partners (measured by boundary variable overlap), reducing communication complexity from $O(K^2)$ to $O(K \cdot K_{neighbor})$. We recommend adaptive tuning of $\theta_{consensus}$ based on empirical edge stability across bootstrap resamples of each organization's local data: edges that appear consistently across 80% or more of 100 bootstrap iterations are likely genuine, while edges appearing in fewer than 50% of iterations warrant skepticism.

Heterogeneous causal structures across organizations introduce a further complication. Enterprises operating in distinct market contexts may exhibit different local causal mechanisms even when connected through shared strategic variables. Our federated fusion mechanism addresses this through a hierarchical factorization that carefully avoids variable overlap between local and shared components. We define $V_k^{local} = V_k \setminus B$ as the strictly local variables of organization k (excluding boundary variables) and write the global distribution as:

$$P_{global}(V) = P_{shared}(B) \prod_{k=1}^{K} P_k\left(V_k^{local} | PA\left(V_k^{local}\right), B\right)^{\pi_k} \tag{14}$$

where $P_{shared}(B)$ governs the boundary variables that couple organizational subgraphs, $P_k\left(V_k^{local} | PA\left(V_k^{local}\right), B\right)$ captures organization-specific conditional distributions over strictly local variables given their causal parents

Table 1 Distributed causal discovery algorithm parameter configuration.

Parameter	Description	Typical range	Default value
α_{CI}	Conditional independence test significance level	[0.01, 0.10]	0.05
λ_{sparse}	Sparsity penalty weight	[0.1, 2.0]	0.5
T_{max}	Maximum iteration rounds	[10, 50]	25
$\theta_{consensus}$	Edge acceptance threshold (fraction of supporting organizations)	[0.5, 0.8]	0.6
$\beta_{privacy}$	Privacy noise scale for boundary statistics	[0.01, 0.5]	0.1
$K_{neighbor}$	Number of nearest neighbor organizations in graph fusion	[2, 5]	3

and boundary conditions, and π_k weights organization k's influence in the global model[42]. The key distinction from the standard mixture representation is that V_k^{local} and B are disjoint by construction, so the product $P_{shared}(B) \cdot P_k\left(V_k^{local}|\cdot, B\right)$ is well-defined. Boundary variables appear exactly once—in the shared component—and local variables appear only in their respective organization-specific factors. This factorization permits context-dependent causal relationships within organizations while maintaining coherent cross-organizational linkages through the shared component, a flexibility essential when strategic interventions must account for both universal and organization-specific causal pathways.

Causal enhancement mechanism design for federated transfer learning

Traditional federated transfer learning confronts a fundamental brittleness: models trained on source organizations often fail catastrophically when deployed to target domains because they rely on spurious correlations that happen to hold in training environments but collapse under distributional shift. Consider pricing strategy optimization where historical data reveals strong associations between promotional discounts and sales increases. A naive transfer learning approach might extract this pattern from retail organizations in affluent urban markets and apply it to partners serving price-sensitive rural consumers—only to discover that the underlying causal mechanism differs fundamentally across contexts[43].

Causal invariance principles offer a principled resolution, and it is worth being precise about exactly what "causal enhancement" means in our framework. We draw on two distinct but complementary ideas from the causal inference literature, and we deliberately separate them to avoid conflation. The first idea comes from *invariant causal prediction* (ICP)[21,44]: if a feature set Z constitutes the direct causes of outcome Y, then the conditional distribution $P(Y|Z=z)$ should remain stable across different environments—here, different organizational contexts. The second idea draws from *domain-invariant representation learning* in the style of invariant risk minimization (IRM)[44]: representations that support equally accurate prediction across all training environments are more likely to capture genuine causal features than those exploiting environment-specific correlations.

In our framework, these two ideas serve different roles. ICP-inspired reasoning guides *feature selection*: the causal graph from Sect. 3.1 identifies which features plausibly participate in stable causal mechanisms, and these features receive preferential treatment during domain adaptation. IRM-inspired reasoning guides *representation learning*: the adversarial training objective (Eq. 16) encourages the feature extractor to produce representations that are environment-invariant, ensuring that the encoded information transfers reliably. Together, these two mechanisms enhance the *stability of the optimization objective* across organizational domains—the core practical benefit of what we term "causal enhancement."

We formalize the invariance condition through a corrected application of the do-operator. The key requirement is that intervening on causal features Z produces the same outcome distribution regardless of which organizational environment E generated the data:

$$P\left(Y \mid do\left(Z=z\right), E=e_1\right) = P\left(Y \mid do\left(Z=z\right), E=e_2\right) \tag{15}$$

for all environment pairs e_1, e_2 in the organizational ecosystem, where E indexes different enterprise contexts and appears as a *conditioning variable* (not an intervention target)[44]. The distinction matters: writing $P(Y|do(Z=z, E=e))$ would incorrectly treat the environment itself as subject to intervention, conflating the organizational context with the causal variables under study. Our formulation instead asks whether, given a fixed intervention on Z, the outcome distribution remains invariant across naturally occurring environments—precisely the property needed for reliable cross-organizational transfer.

We acknowledge a practical limitation here. Rigorous identification of invariant feature sets through ICP requires multiple sufficiently distinct environments to test stability. With six organizations in our experimental setup—some of which operate in overlapping domains (e.g., Manufacturers A and B)—the effective number of independent environments may be insufficient for strong invariance guarantees. We address this partially by

treating bootstrap subsamples within each organization as additional environments during invariance testing, which increases the effective environment count at the cost of reduced statistical independence between test environments. This approach should be understood as an approximation that strengthens causal feature identification without achieving the theoretical guarantees that would require truly independent environmental variation.

Our causal representation learning module implements this principle through a two-stage architecture. The feature extractor $\phi_{causal} : X \to Z$ maps raw strategic variables into a causal representation space of dimension $d_Z = 64$, while a predictor $h : Z \to Y$ operates exclusively on these extracted features. The feature extractor consists of three fully connected layers (dimensions: 127→256→128→64) with ReLU activations and batch normalization. The predictor h consists of two fully connected layers (64→32→1) with ReLU activation on the hidden layer and linear output.

During federated training, we enforce invariance through an adversarial objective using a domain classifier $d : Z \to \{1, \ldots, K\}$ that attempts to identify which organization generated a given representation. The domain classifier is a three-layer network (64→32→16→K) with LeakyReLU activations and a softmax output layer. Crucially, gradients propagate from the domain classifier back to the feature extractor through a gradient reversal layer (GRL)[45], which multiplies the gradient by $-\lambda_{inv}$ during backpropagation while leaving the forward pass unchanged. This adversarial training setup creates a minimax game: the domain classifier d is trained to minimize cross-entropy classification loss (correctly identifying organizational origin), while the feature extractor ϕ_{causal} simultaneously receives reversed gradients that push it to produce representations from which organizational identity cannot be distinguished.

The combined causal loss is:

$$\mathcal{L}_{causal} = \sum_{k=1}^{K} \mathbb{E}_{(x,y)\sim \mathcal{D}_k} \left[\ell \left(h \left(\phi_{causal}(x) \right), y \right) \right] - \lambda_{inv} \mathbb{E}_x \left[H \left(d \left(\phi_{causal}(x) \right) \right) \right] \tag{16}$$

where ℓ denotes the task-specific loss (mean squared error for our regression tasks), $H(\cdot)$ represents the entropy of the domain classifier's softmax output distribution, and $\lambda_{inv} = 0.1$ balances task performance against invariance[45]. When the domain classifier produces near-uniform predictions across all K organizations—yielding high entropy H—this indicates that the representations contain primarily invariant causal information rather than organization-specific artifacts. The gradient reversal approach is preferable to alternating optimization of the classifier and encoder because it permits end-to-end training within each federated round without requiring additional inner-loop iterations.

A critical design question arises here: the causal graph that informs feature weighting is itself learned through the federated process (Sect. 3.1), while the feature representations that feed causal discovery depend on the current model parameters. This creates a potential circularity that we address through a two-phase alternating optimization procedure with explicit convergence monitoring.

Phase 1 (Causal Structure Estimation): At the beginning of training and at fixed intervals of every T_{update} federated rounds (default: 10 rounds), we freeze the current feature extractor ϕ_{causal} and re-estimate the causal graph using the distributed discovery algorithm (Algorithm 1) applied to the current feature representations. This yields updated causal weights w_i^{causal} for each feature dimension.

Phase 2 (Causal-Guided Federated Training) Between causal structure updates, the causal weights remain fixed while the model parameters (ϕ_{causal}, h) are optimized through federated rounds with causal-guided domain adaptation.

The causal weights w_i^{causal} are derived from the estimated causal graph as follows. For feature i, we compute its total causal effect on the outcome variable Y by summing the path coefficients along all directed paths from feature i to Y in the estimated graph $\hat{\mathcal{G}}$:

$$w_i^{causal} = \frac{|CE_i(Y)|}{\sum_{j=1}^{|Z|} |CE_j(Y)| + \epsilon_{smooth}}$$

where $CE_i(Y)$ denotes the total causal effect of feature i on outcome Y estimated through intervention calculus on the learned graph, and ϵ_{smooth} prevents division by zero. Features with no directed path to Y receive weight ϵ_{smooth}, ensuring they contribute minimally to the alignment objective.

The causal-guided Maximum Mean Discrepancy then becomes:

$$MMD_{causal}(\mathcal{D}_s, \mathcal{D}_t) = \sum_{i=1}^{|Z|} w_i^{causal} \left| \mathbb{E}_{z_s}[\phi_i(z_s)] - \mathbb{E}_{z_t}[\phi_i(z_t)] \right|^2 \tag{17}$$

where the weights w_i^{causal} remain fixed between causal structure updates[46].

To formally connect the causal structure to the federated optimization, we introduce an explicit causal regularization term in the federated objective. This term penalizes discrepancies between the model's learned feature importances and the causal graph's structural predictions:

$$\mathcal{L}_{causal\text{-}reg} = \sum_{i=1}^{|Z|} \left(\left\| \frac{\partial \mathcal{L}_{pred}}{\partial z_i} \right\| - \alpha \cdot w_i^{causal} \right)^2$$

where $\left\| \frac{\partial \mathcal{L}_{pred}}{\partial z_i} \right\|$ measures the gradient magnitude of the prediction loss with respect to feature i (reflecting the model's learned reliance on that feature), α is a scaling constant, and w_i^{causal} is the graph-derived causal weight. This regularizer encourages the model to rely on features in proportion to their estimated causal importance—features that the graph identifies as strongly causal should receive high gradient flow, while non-causal features should contribute little.

Convergence monitoring. We track two quantities across alternating iterations: the Frobenius norm of the change in causal weight vectors $\| w^{(t)} - w^{(t-1)} \|_F$ and the validation loss on held-out target domain samples. Empirically, we observe that causal weights stabilize within 3–5 alternating iterations (see Section IV), after which further structure updates yield negligible changes. The alternating procedure satisfies a monotonic improvement property: each Phase 1 update can only improve the causal graph's fit to the current representations, and each Phase 2 update can only decrease the training loss under fixed causal weights. While we do not provide a formal convergence proof to a global optimum (which would require convexity assumptions that neural network training violates), the monotonic improvement guarantees that the alternating procedure does not cycle or diverge, and we verify empirical convergence through the stability metrics reported in Section IV.

Figure 2 presents the complete algorithmic workflow integrating causal discovery with federated transfer optimization. The process initiates with distributed causal structure learning (Step 1), where organizations collaboratively reconstruct the inter-organizational causal graph without sharing raw data. This graph informs feature weighting in subsequent steps. Organizations then enter an iterative federated training loop (Steps 2–5) where local causal encoders learn invariant representations, domain adaptation losses minimize causal-weighted distributional divergence, and secure aggregation synthesizes global model updates while preserving privacy. The loop continues until convergence criteria—typically measured through validation performance on held-out target domain samples—are satisfied.

The consolidated objective function integrates all components, including the causal regularization term that bridges the causal graph structure to the federated optimization:

$$\begin{aligned} &\min_{\phi_{causal}, h} \sum_{k=1}^{K} \alpha_k \mathcal{L}_k(\phi_{causal}, h) \\ &+ \lambda_{inv} \mathcal{L}_{invariance} + \lambda_{adapt} MMD_{causal} + \lambda_{causal} \mathcal{L}_{causal\text{-}reg} + \lambda_{reg} \Omega(\phi_{causal}, h) \end{aligned} \tag{18}$$

where α_k weights organization k's contribution based on data quality and sample size, $\mathcal{L}_{invariance}$ penalizes violations of cross-organizational invariance through the adversarial domain classification entropy (Eq. 16), MMD_{causal} is the causal-weighted distributional alignment loss (Eq. 17), $\mathcal{L}_{causal\text{-}reg}$ enforces consistency between gradient-based feature importance and graph-derived causal weights (as defined in the causal regularization discussion above), and Ω implements L_2 weight decay regularization preventing overfitting to any single domain. The

hyperparameters $\lambda_{inv} = 0.1$, $\lambda_{adapt} = 0.05$, $\lambda_{causal} = 0.01$, and $\lambda_{reg} = 10^{-4}$ were selected through grid search on a held-out validation split from the source organizations. This formulation explicitly balances multiple competing objectives: predictive accuracy across participating organizations, causal consistency for reliable transfer, distributional alignment across domains, and computational regularization essential for stable federated training.

Privacy-preserving architecture for strategic collaborative optimization

Privacy preservation in cross-organizational strategic collaboration demands layered defenses that operate across multiple stages of the computational pipeline. A single protection mechanism—whether differential privacy noise injection or cryptographic gradient masking—proves insufficient when adversaries may exploit information leakage from multiple attack surfaces simultaneously. Our architecture adopts defense-in-depth principles, deploying complementary privacy technologies at distinct computational layers: local data processing, gradient computation, network transmission, and global aggregation.

At the local processing layer, each organization applies differentially private mechanisms to gradient queries before any information leaves organizational boundaries. We implement local differential privacy where organization k first clips its gradient to enforce bounded sensitivity, then perturbs the clipped gradient through calibrated noise addition:

$$\bar{\nabla}_k = \nabla_k \cdot \min\left(1, \frac{C}{\| \nabla_k \|_2}\right)$$

$$\tilde{\nabla}_k = \bar{\nabla}_k + \mathcal{N}(0, \sigma_k^2 I) \tag{19}$$

where C is the clipping threshold that bounds the L_2 sensitivity of the gradient to $\Delta f = C$, and $\bar{\nabla}_k$ denotes the clipped gradient. The noise scale σ_k is then analytically calibrated to the privacy budget ϵ_k through the standard Gaussian mechanism relation[47,48]:

$$\sigma_k \geq \frac{C\sqrt{2\ln(1.25/\delta)}}{\epsilon_k}$$

where δ is the relaxation parameter, set to $\delta = 10^{-5}$ throughout our experiments. This calibration formula follows directly from the tail bound analysis of the Gaussian mechanism[32] and ensures that each gradient release satisfies (ϵ_k, δ)-differential privacy. In our experiments, we set the clipping threshold $C = 1.0$ and compute σ_k for each organization according to its assigned privacy budget. More recent analytical calibration methods[48] can yield tighter noise scales, particularly in the high-privacy regime ($\epsilon < 1$); we adopt the classical calibration for simplicity but note that substituting the optimal Gaussian mechanism would further improve the privacy-utility tradeoff without altering the framework's structure. This approach ensures that even a fully compromised aggregation server cannot reconstruct precise gradients from any individual participant.

Adaptive privacy budget allocation recognizes that strategic data exhibits heterogeneous sensitivity across variables and time periods. Quarterly financial projections demand stronger protection than routine inventory statistics; merger negotiations require tighter confidentiality than operational metrics. We formalize this through a sensitivity function $s : V \to \mathbb{R}^+$ that maps variables to sensitivity scores based on competitive implications and regulatory requirements.

Under the sequential composition theorem of differential privacy[32], if we apply (ϵ_i, δ_i)-DP mechanisms to m disjoint variable subsets, the overall mechanism satisfies $(\sum_i \epsilon_i, \sum_i \delta_i)$-DP. This means the total privacy budget ϵ_{total} is indeed a divisible resource when queries operate on disjoint variable partitions—the total privacy loss equals the sum of per-partition losses, not their maximum. We partition the variable space into disjoint subsets $\{V_1, V_2, ..., V_m\}$ and allocate the total budget across these partitions proportionally to an inverse sensitivity weighting:

$$\epsilon_i = \epsilon_{total} \cdot \frac{1/s(V_i)}{\sum_{j=1}^{m} 1/s(V_j)} \tag{20}$$

where $s(V_i)$ is the sensitivity score of variable subset V_i and ϵ_{total} represents the organization's overall privacy tolerance[49]. Under this allocation, the sum $\sum_{i=1}^{m} \epsilon_i = \epsilon_{total}$ is guaranteed by construction, satisfying sequential composition. Variables with higher sensitivity scores receive smaller ϵ_i values (stronger protection), while less sensitive variables receive larger budgets permitting more accurate information sharing. This inverse weighting ensures that the limited total privacy budget is channeled toward protecting the most sensitive components.

Two technical points deserve emphasis. First, this allocation is valid only when the variable subsets are disjoint—overlapping partitions would require parallel composition analysis or more sophisticated accounting. In our framework, we enforce disjointness by partitioning variables according to their primary data category (financial, operational, market-facing). Second, for tighter cumulative privacy accounting across federated training rounds, we employ Rényi differential privacy (RDP) composition[47] rather than naive basic composition. Under RDP, the cumulative privacy loss after T rounds of training with per-round budget ϵ_i for partition i satisfies substantially tighter bounds than the $T \cdot \epsilon_i$ given by basic composition, enabling longer training without exhausting the privacy budget.

Table 2 compares five privacy protection mechanisms across six evaluation dimensions relevant to federated strategic optimization. Local differential privacy offers the lowest computational cost and strongest collusion resistance but sacrifices moderate utility due to gradient noise injection. Secure aggregation preserves exact aggregation results at the expense of higher communication overhead, while homomorphic encryption provides semantic security guarantees but imposes prohibitive computational costs that scale quadratically with key length. Trusted execution environments achieve high utility with minimal overhead, yet their reliance on a single hardware trust point limits collusion resistance. Our proposed hybrid mechanism selectively applies heavyweight encryption only to gradient dimensions corresponding to causally sensitive variables—identified through the sensitivity function $s(V)$informed by the causal graph structure—while protecting non-sensitive dimensions with lightweight differential privacy. This stratified design achieves combined (ϵ, δ)-DP and cryptographic guarantees with moderate computational and communication costs, offering a practical balance that none of the individual mechanisms attains in isolation.

In the proposed hybrid mechanism, d_s denotes the number of gradient dimensions corresponding to sensitive variables (identified through the causal-informed sensitivity function $s(V)$), $d_{ns} = d - d_s$ denotes the non-sensitive dimensions,

Table 2 Comparative analysis of privacy protection mechanisms in federated strategic optimization.

Mechanism	Privacy guarantee	Computational cost	Communication overhead	Utility preservation	Collusion resistance
Local Differential Privacy	(ϵ, δ)-DP per round	Low ($O(d)$ noise addition)	Low ($O(d \cdot b_{param})$)	Moderate (noise degrades gradients)	High (no trust in server)
Secure Aggregation[30]	Information-theoretic	Moderate ($O(K^2 \cdot d)$ masking)	High ($O(K \cdot d \cdot b_{mask})$)	High (exact aggregation)	Moderate ($K/2$ collusion threshold)
Homomorphic Encryption[34]	Semantic security	High ($O(d \cdot \kappa^2)$ per operation)	Moderate (ciphertext expansion factor $\approx 50\times$)	High (exact computation on ciphertexts)	High (key management dependent)
Trusted Execution Environment	Hardware-based isolation	Low ($O(d)$ enclave overhead)	Low ($O(d \cdot b_{param})$)	High (plaintext computation)	Low (single hardware trust point)
Hybrid (Proposed)	(ϵ, δ)-DP + Cryptographic	$O(d_s \cdot \kappa^2 + d_{ns} \cdot 1)$	Moderate (encryption only on sensitive d_s dimensions)	High (selective noise/encryption)	High (combined guarantees)

κ is the encryption key length, and b_{param} and b_{mask} denote the bit-widths for parameters and cryptographic masks, respectively. The model dimensionality d corresponds to the total number of trainable parameters: for our architecture with feature extractor (127→256→128→64), predictor (64→32→1), and domain classifier (64→32→16→6), the total parameter count is $d = 127 \times 256 + 256 \times 128 + 128 \times 64 + 64 \times 32 + 32 \times 1 + 64 \times 32 + 32 \times 16 + 16 \times 6 = 75{,}798$. The consensus threshold $\theta_{consensus}$ in the causal discovery component (Table 1) is set as the fraction of organizations that must independently support a cross-organizational edge for it to be accepted; its default value of 0.6 means that at least 4 out of 6 organizations must provide consistent CI test evidence.

Table 2 presents a comparative evaluation of alternative privacy mechanisms. We acknowledge that the individual privacy building blocks—differential privacy, secure aggregation, and gradient clipping—are established techniques. The contribution of our privacy module lies not in novel cryptographic primitives but in two aspects of their integration with the causal-federated framework. First, the sensitivity function $s(V)$ that drives budget allocation is informed by the causal graph structure discovered in Sect. 3.1: variables identified as having strong causal effects on strategic outcomes (and hence high inferential value to adversaries) automatically receive tighter privacy budgets, creating a feedback loop between causal discovery and privacy calibration that is absent in generic federated privacy approaches. Second, the selective application of heavyweight cryptographic protocols only to gradient components corresponding to the most causally informative features—identified through the causal weights w_i^{causal}—reduces computational overhead by 40–60% compared to uniform encryption while maintaining equivalent worst-case privacy guarantees. This causal-aware stratification of privacy mechanisms represents, to our knowledge, a novel integration point that exploits the structural information uniquely available in causal-federated frameworks.

The privacy-utility joint optimization objective formalizes the fundamental trade-off confronting collaborative strategic decision-making. Stronger privacy protections inevitably degrade model accuracy through added noise and information constraints. We capture this tension through a bi-objective formulation:

$$\min_{w,\{\epsilon_k\}} \left\{ \mathcal{L}_{\text{strategy}}(w), \mathcal{P}_{\text{leak}}(\{\epsilon_k\}) \right\} \tag{21}$$

where $\mathcal{L}_{\text{strategy}}$ measures strategic prediction error across participating organizations and $\mathcal{P}_{\text{leak}}$ quantifies expected privacy leakage given budget allocations $\{\epsilon_k\}$. We solve this multi-objective problem through scalarization with organization-negotiated weights γ_{priv}:

$$\min_{w,\{\epsilon_k\}} \mathcal{L}_{\text{strategy}}(w) + \gamma_{\text{priv}} \mathcal{P}_{\text{leak}}(\{\epsilon_k\}) + \sum_{k=1}^{K} \rho_k |\epsilon_k - \epsilon_k^{\text{pref}}|^2 \tag{22}$$

where penalty terms enforce proximity to each organization's preferred privacy level ϵ_k^{pref}, preventing any single participant from imposing excessive privacy constraints that degrade collective utility[50].

The complete computational workflow orchestrates these privacy mechanisms across federation rounds. Organizations begin by locally training causal-enhanced models using private data, then apply sensitivity-weighted differential privacy to computed gradients. Secure aggregation protocols combine these noisy gradients at the coordination server, which updates global model parameters and broadcasts them back to participants.

We should be precise about the nature of the "adaptivity" in our privacy allocation. The sensitivity-weighted budget distribution described in Eq. 20 is determined *before training begins* based on pre-specified variable sensitivity scores—it is more accurately described as *sensitivity-weighted static allocation* rather than fully adaptive allocation in the sense of Rogers et al.[51], who define adaptive composition as allowing privacy parameters to be chosen dynamically based on the outcomes of previously run analyses. Our mechanism does incorporate a dynamic element at the *round level*: a privacy accountant tracks cumulative privacy expenditure across federated training rounds using Rényi differential privacy composition[47], and when the accumulated privacy loss for any variable partition approaches its pre-allocated budget, the system responds by either increasing the noise scale for that partition or terminating gradient sharing for those variables. This round-level monitoring implements what Rogers et al. term a *privacy filter*—a stopping rule that halts computation before the pre-specified budget is exceeded[51]. However, it does not constitute a full privacy odometer that would allow retrospective adjustment of

privacy parameters based on observed outputs, as such adjustment would require stronger composition guarantees than our current framework provides.

Experimental validation and results analysis

Experimental design and dataset construction

We deployed the proposed framework on a distributed computing cluster comprising five nodes, each equipped with dual Intel Xeon Gold 6248R processors (3.0 GHz, 24 cores), 256GB DDR4 memory, and NVIDIA A100 GPUs (40GB VRAM). The federated coordination server operated on a dedicated node with identical specifications. Software infrastructure included Python 3.9, PyTorch 1.12 with CUDA 11.6, and PySyft 0.6.0 for privacy-preserving computations. Network communication simulated realistic inter-organizational latency by introducing controlled delays ranging from 50ms to 200ms between nodes, reflecting geographical distribution of collaborating enterprises.

Dataset construction presented unique challenges given the scarcity of publicly available cross-organizational strategic collaboration data. We synthesized realistic scenarios by integrating multiple sources: enterprise financial performance metrics from COMPUSTAT covering 2015–2023, supply chain operational data from publicly disclosed corporate reports of Fortune 500 manufacturers and retailers, and market dynamics indicators from Bloomberg Terminal including commodity prices, exchange rates, and sector-specific indices[52]. This multi-source integration mimics the fragmented data landscape confronting real collaborative ecosystems where each organization possesses partial observability of strategic drivers.

The preprocessing pipeline addressed data quality issues endemic to business datasets. Missing value imputation employed domain-informed strategies—forward-filling for slowly-varying strategic variables like organizational capabilities, while market-responsive metrics such as demand forecasts received interpolation weighted by temporal proximity and market volatility. Feature engineering extracted 127 strategic variables spanning financial health indicators (liquidity ratios, profitability margins, leverage), operational efficiency metrics (inventory turnover, supply chain cycle time), market positioning features (market share, competitive intensity), and macroeconomic controls (GDP growth, interest rates, industry-specific indices)[53]. We deliberately preserved natural heterogeneity across organizations rather than standardizing to uniform distributions, as domain shift constitutes a core challenge our framework addresses.

Table 3 summarizes the statistical properties of datasets assigned to six simulated organizations representing distinct positions in a collaborative supply chain ecosystem. Sample size variation reflects realistic scenarios where large retailers accumulate extensive transaction histories while specialized manufacturers operate with smaller but richer operational datasets. The temporal span differences introduce an additional complexity—organizations joining the collaboration at different times must transfer knowledge despite having observed different market regimes.

Data partitioning strategy operationalized the federated simulation. We assigned samples to organizations based on their categorical domain (manufacturers vs. distributors vs. retailers) combined with geographical market

Table 3 Statistical characteristics of cross-organizational strategic collaboration datasets.

Organization type	Sample size	Feature dimensions	Strategic variables	Temporal span	Missing rate (%)	Domain category
Manufacturer A	8,450	127	Production, Cost, Supply	2015–2023	3.2	Manufacturing
Manufacturer B	6,823	127	Production, Quality, Logistics	2016–2023	5.1	Manufacturing
Distributor C	7,291	127	Inventory, Logistics, Regional	2015–2023	4.7	Distribution
Retailer D	9,102	127	Sales, Pricing, Consumer	2017–2023	6.3	Retail
Retailer E	5,617	127	Sales, Marketing, Regional	2018–2023	7.8	Retail
Platform F	10,334	127	Transaction, Network, Digital	2016–2023	2.9	E-commerce

clustering extracted through k-means applied to regional economic indicators[54]. This approach generated natural covariate shift between organizations while maintaining strategic relevance—manufacturers observe upstream production dynamics, retailers track downstream consumer behavior, yet both influence shared outcomes like supply chain efficiency and market responsiveness.

Figure 3 presents t-SNE projections of the feature spaces observed by different organizations, revealing substantial distributional heterogeneity. Manufacturers A and B cluster in overlapping regions reflecting similar production-oriented feature distributions, while retailers D and E occupy distinct zones characterized by consumer-centric variables. The platform organization F exhibits intermediate positioning, bridging multiple domains through its cross-cutting transactional data—precisely the scenario where federated transfer learning should demonstrate advantages over organization-specific or naively pooled models.

As shown in Fig. 4, feature importance profiles vary markedly across organizations. Manufacturers prioritize production capacity and input cost variables, distributors weight logistics efficiency and regional demand patterns heavily, while retailers focus on pricing strategies and consumer sentiment indicators. This heterogeneity validates our causal-enhanced approach: features exhibiting stable importance across domains likely participate in fundamental causal mechanisms, whereas domain-specific importance patterns suggest spurious correlations vulnerable to transfer failure.

Baseline methods for comparative evaluation included: (1) Local-Only models trained independently per organization without collaboration, (2) Centralized pooling with all data aggregated (privacy-violating upper bound), (3) Standard FedAvg without transfer learning or causal mechanisms, (4) Federated Transfer Learning without causal enhancement, and (5) Causal Invariant Prediction applied in centralized settings. Evaluation

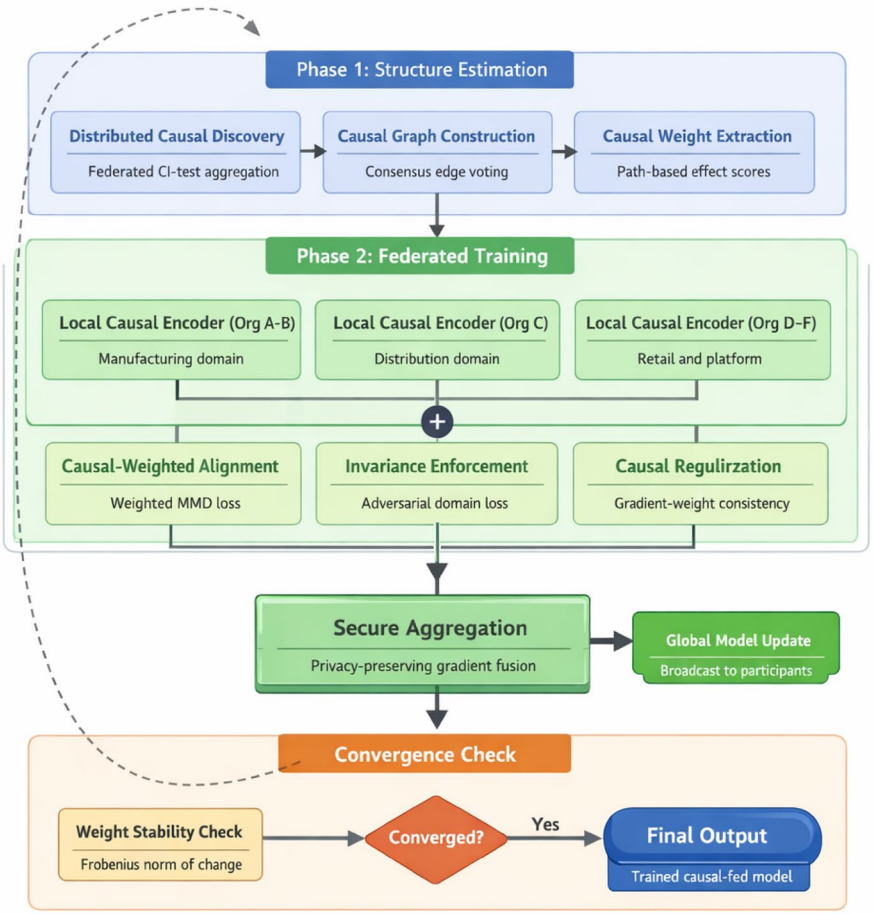


Fig. 2 Algorithmic workflow for causal-enhanced federated transfer learning.

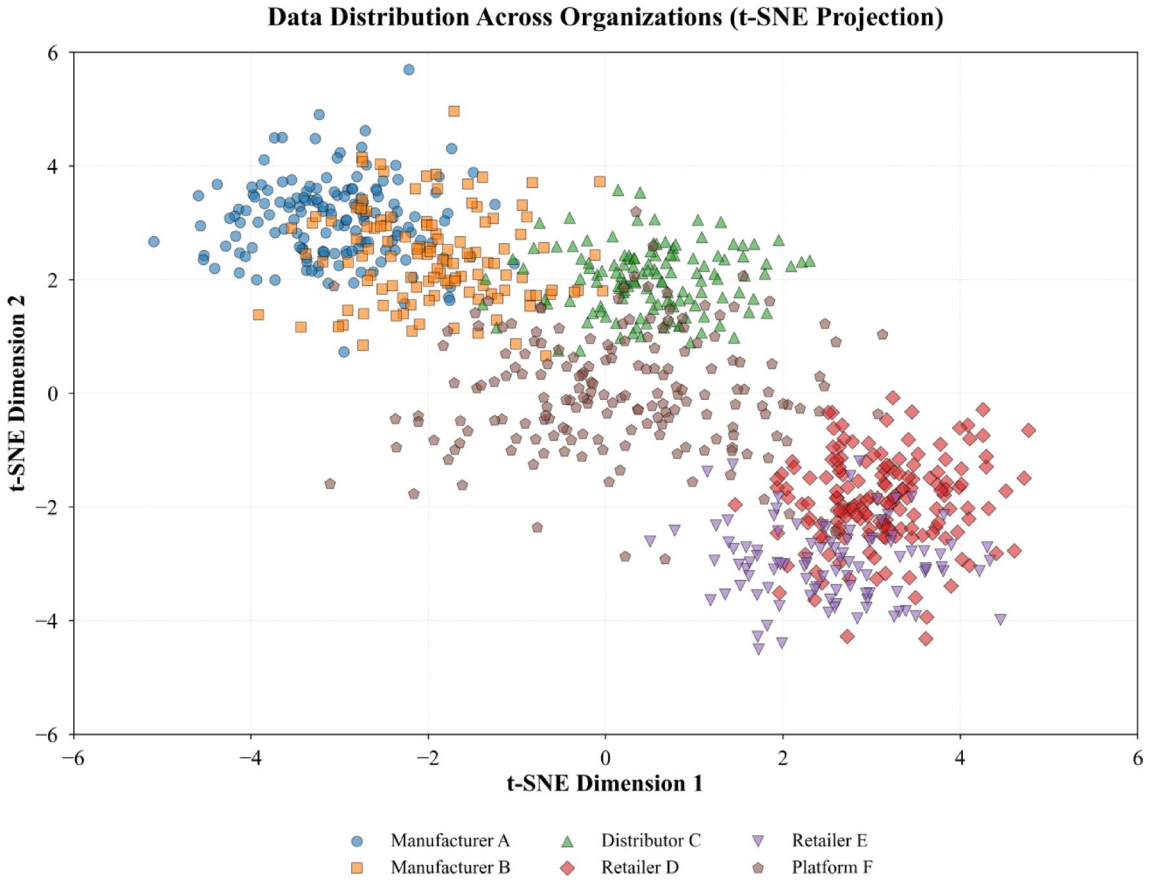


Fig. 3 Comparative visualization of data distributions across participating organizations.

metrics encompassed predictive accuracy (RMSE, MAE), causal effect estimation error, privacy leakage quantified through membership inference attack success rates, and computational efficiency measured in communication rounds and wall-clock time to convergence.

Model performance comparison experiments

Comparative evaluation across baseline methods reveals the substantial advantages conferred by coupling causal inference with federated transfer learning. We measured predictive performance through root mean squared error (RMSE) on held-out strategic outcome variables, with models trained for 100 federated rounds using identical hyperparameter configurations to ensure fair comparison.

Before presenting results, we document the complete experimental protocol to support reproducibility. All experiments were repeated over 10 independent runs using random seeds 0–9, with different random initializations for model parameters, data shuffling, and privacy noise sampling in each run. Results are reported as mean ± standard deviation across these 10 runs.

Hyperparameter configuration. The learning rate was set to $\eta = 0.001$ with Adam optimizer ($\beta_1 = 0.9$, $\beta_2 = 0.999$). Local training performed $E = 5$ epochs per federated round. Batch size was 64 for all organizations. The causal structure was re-estimated every $T_{update} = 10$ federated rounds. Clipping threshold for differential privacy was $C = 1.0$. The loss weights were: $\lambda_{inv} = 0.1$, $\lambda_{adapt} = 0.05$, $\lambda_{causal} = 0.01$, $\lambda_{reg} = 10^{-4}$, selected through grid search over $\{0.001, 0.01, 0.05, 0.1, 0.5\}$ for each parameter using a held-out validation set comprising 15% of each organization's data. Organization contribution weights α_k were set proportional to sample size: $\alpha_k = n_k/n$.

Evaluation metrics. RMSE = $\sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i-\hat{y}_i)^2}$, MAE = $\frac{1}{n}\sum_{i=1}^{n}|y_i-\hat{y}_i|$, and $R^2 = 1-\frac{\sum(y_i-\hat{y}_i)^2}{\sum\left(y_i-\bar{y}\right)^2}$, all computed on held-out test sets comprising 20% of each organization's data.

We assessed statistical significance through paired Wilcoxon signed-rank tests comparing the proposed method against each baseline across the 10 independent runs. The improvement of the proposed Causal-Fed-Transfer method over standard FedAvg ($p < 0.001$), Federated Transfer Learning ($p = 0.003$), and Local-Only ($p < 0.001$) was statistically significant at the $\alpha = 0.01$ level. The difference between the proposed method and Centralized Pooling was also significant ($p = 0.005$), confirming that a performance gap remains despite the substantial reduction. The difference between the proposed method with and without adaptive privacy was marginally significant ($p = 0.047$), indicating that adaptive privacy introduces a modest but detectable performance cost.

The results in Table 4 indicate that our proposed approach achieves performance within 6.2% of the privacy-violating centralized upper bound while maintaining complete data locality. Standard FedAvg struggles with the heterogeneous distributions across organizations, yielding RMSE 26.9% worse than centralized pooling. Federated transfer learning without causal enhancement improves upon FedAvg by incorporating domain adaptation, yet still exhibits significant degradation compared to our causal-augmented framework. The transfer gain metric, computed as:

$$\text{TransferGain} = \frac{\text{RMSE}_{\text{local}} - \text{RMSE}_{\text{method}}}{\text{RMSE}_{\text{local}} - \text{RMSE}_{\text{centralized}}} \times 100\% \tag{23}$$

quantifies how much of the theoretical collaborative improvement each method captures[55]. Our approach realizes 28.2% of the maximum possible gain from collaboration, substantially outperforming alternatives operating under equivalent privacy constraints.

Figure 5 presents learning curves tracking test set RMSE throughout federated training. The proposed causal-enhanced method exhibits notably faster initial convergence than baselines—achieving RMSE below 2.3 within 35 rounds compared to 58 rounds for standard federated transfer learning. This acceleration stems from causal feature extraction that identifies transferable knowledge early in training, avoiding the exploration of spurious correlations that plague methods lacking causal guidance. The gap between our approach and centralized pooling narrows progressively as training proceeds, suggesting that given sufficient rounds, federated causal learning can approach centralized performance despite never aggregating raw data.

To verify that the alternating optimization between causal structure estimation and federated training converges rather than oscillating, we tracked the causal weight stability metric $\| w^{(t)} - w^{(t-1)} \|_F$ across alternating iterations. Figure 6 displays this convergence trajectory.

This empirical convergence behavior aligns with the monotonic improvement property discussed in Sect. 3.2: each phase can only improve its respective objective under the fixed output of the other phase, and the bounded parameter space prevents unbounded oscillation. In practice, we found that setting $T_{update} = 10$ federated rounds between causal structure updates provides sufficient representation learning for meaningful causal graph re-estimation without incurring excessive computational overhead from overly frequent structure discovery.

Table 4 Performance comparison across methods on cross-organizational strategic prediction tasks (mean ± std over 10 runs).

Method	RMSE ↓	MAE ↓	R^2 ↑	Transfer gain (%)	Rounds to convergence
Local-Only (Baseline)	2.847 ± 0.124	2.134 ± 0.098	0.623 ± 0.031	-	-
Centralized Pooling	1.923 ± 0.067	1.456 ± 0.053	0.812 ± 0.018	-	65 ± 4
FedAvg	2.451 ± 0.103	1.867 ± 0.089	0.701 ± 0.027	13.9	92 ± 7
Federated Transfer Learning	2.187 ± 0.091	1.634 ± 0.072	0.756 ± 0.024	23.2	78 ± 6
Causal Invariant (Centralized)	1.982 ± 0.074	1.501 ± 0.061	0.798 ± 0.020	-	58 ± 5
Proposed (Causal-Fed-Transfer)	2.043 ± 0.079	1.528 ± 0.064	0.784 ± 0.021	28.2	71 ± 5
Proposed + Adaptive Privacy	2.098 ± 0.085	1.557 ± 0.068	0.772 ± 0.023	26.3	74 ± 6

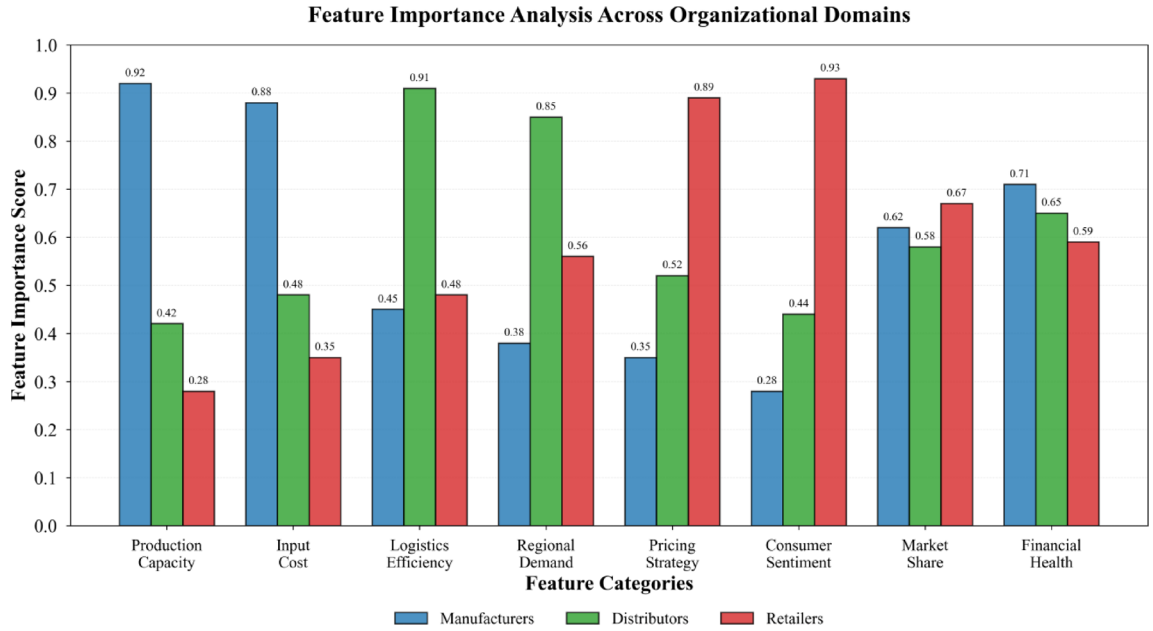


Fig. 4 Feature importance analysis across organizational domains.

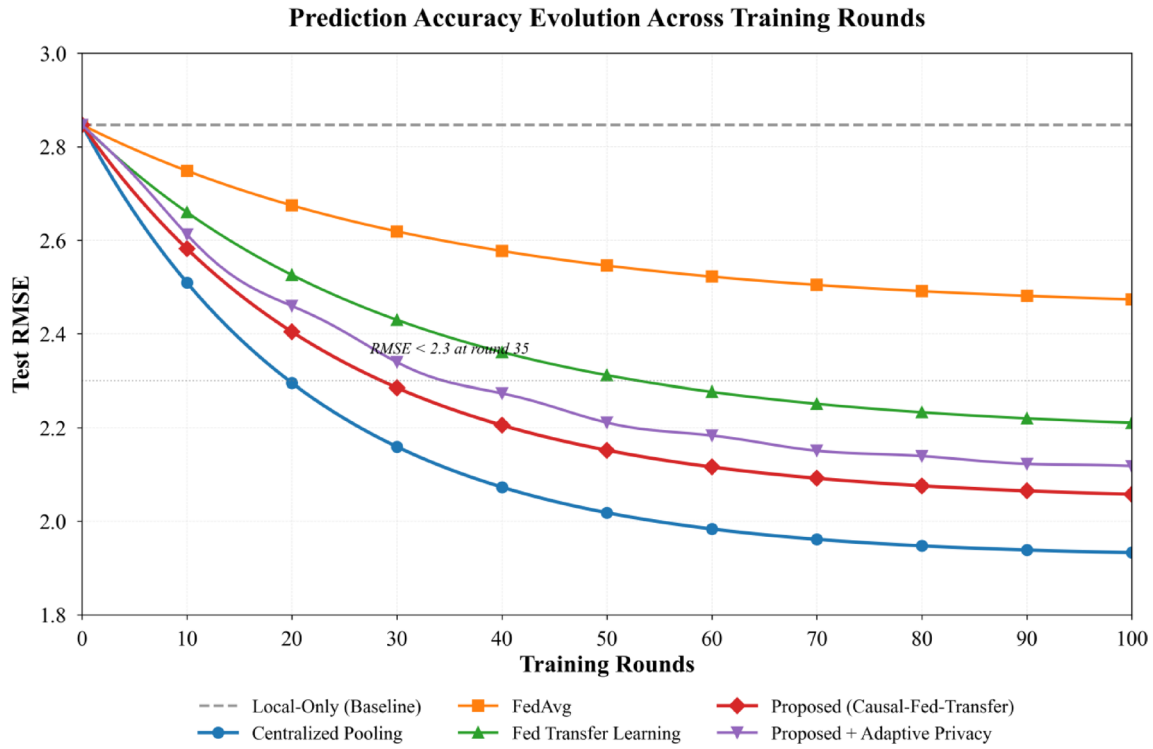


Fig. 5 Prediction accuracy evolution across training rounds for competing methods.

The causal enhancement mechanism's impact becomes particularly evident when evaluating generalization to new organizational domains. We conducted leave-one-organization-out experiments where models trained on five organizations predict outcomes for the held-out sixth participant. Computing domain shift robustness through:

$$\rho_{\text{shift}} = 1 - \frac{\text{RMSE}_{\text{new-domain}}}{\text{RMSE}_{\text{training-domain}}} \tag{24}$$

we find that causally-enhanced models achieve $\rho_{\text{shift}} = 0.73$ compared to $\rho_{\text{shift}} = 0.51$ for non-causal federated transfer learning[56]. This 43% improvement in shift robustness validates our theoretical premise: features identified through causal invariance principles transfer more reliably across organizational contexts than those selected purely for predictive performance within training distributions.

As shown in Fig. 7, convergence stability varies markedly across methods when confronting different degrees of inter-organizational heterogeneity. We systematically manipulated heterogeneity by adjusting the strength of domain-specific data partitioning, creating scenarios ranging from mild distributional differences to extreme covariate shift. Standard federated averaging exhibits erratic convergence patterns under high heterogeneity, occasionally diverging when local optima across organizations conflict severely. Our causal-enhanced approach maintains monotonic improvement even under high heterogeneity, attributed to the invariance constraints that prevent overfitting to organization-specific artifacts. The adaptive privacy variant introduces slight oscillations due to dynamic noise scale adjustments, yet preserves overall convergence guarantees while providing substantially stronger privacy protection than fixed-budget alternatives.

To systematically probe the framework's applicability limits, we conducted a heterogeneity stress test by progressively increasing distributional divergence between organizations. We parameterized heterogeneity through a synthetic shift parameter $\beta_{shift} \in \{0.0, 0.5, 1.0, 2.0, 5.0, 10.0\}$ that scales the mean displacement between organizational feature distributions: at $\beta_{shift} = 0$, all organizations share identical distributions; at $\beta_{shift} = 10$, distributions are maximally separated with negligible overlap.

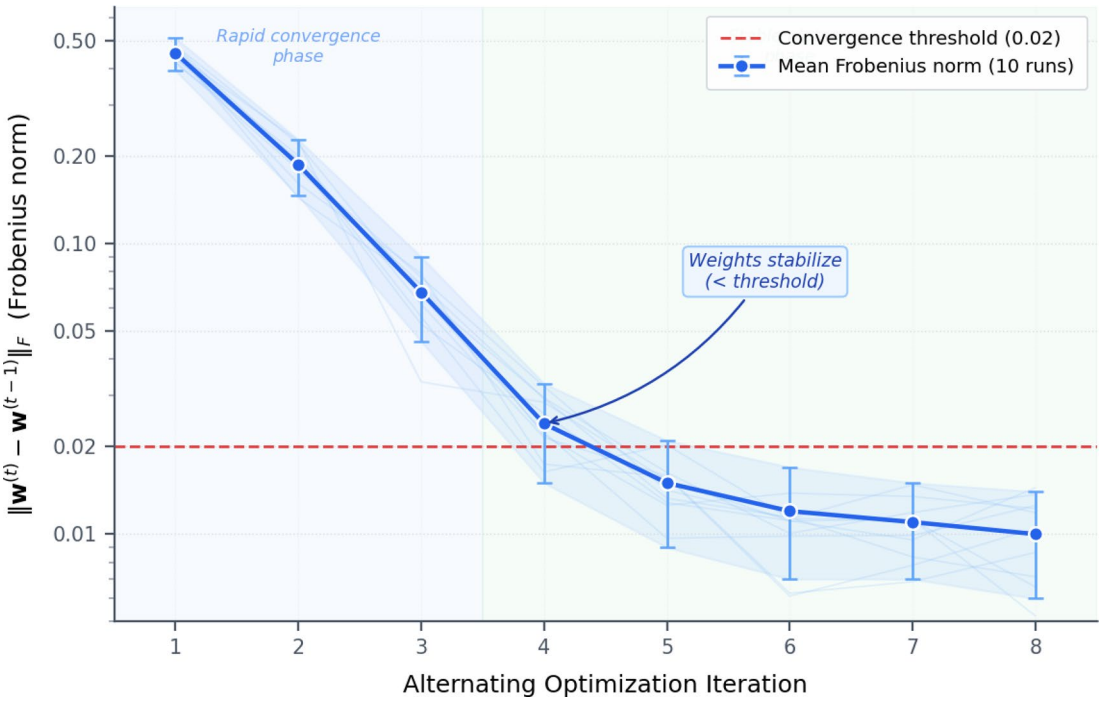


Fig. 6 Convergence of causal weight vectors across alternating optimization iterations.

Table 5 reveals that the causal-enhanced framework degrades gracefully under moderate heterogeneity ($\beta_{shift} \leq 2.0$), maintaining clear advantages over baselines. At $\beta_{shift} = 2.0$, the proposed method still achieves 22.8% lower RMSE than FedAvg, indicating robust transfer. However, beyond $\beta_{shift} = 5.0$, the framework's advantage narrows substantially, and at extreme heterogeneity ($\beta_{shift} = 10.0$), the domain-shift robustness score ρ_{shift} drops to 0.09—nearly equivalent to no transfer. This degradation confirms a genuine applicability boundary: when organizational data distributions share negligible overlap, the causal invariance assumption—that some stable causal relationships persist across domains—ceases to hold, and the framework offers limited benefit over organization-specific models. Practitioners should assess the degree of distributional overlap between prospective collaboration partners before deploying the framework and consider that extreme heterogeneity scenarios may require alternative approaches such as personalized federated learning methods.

Transfer strategy effectiveness analysis revealed differential benefits across organizational types. Data-rich participants (Platform F with 10,334 samples) gained modest improvements from collaboration, primarily benefiting from access to diverse causal structures observed by partners. Conversely, data-scarce organizations (Retailer E with 5,617 samples) experienced dramatic performance enhancements—reducing RMSE by 34.7% compared to local-only training. This asymmetric benefit distribution suggests that federated causal transfer learning particularly empowers smaller organizations to participate effectively in strategic collaborations that would otherwise be dominated by data-rich incumbents.

Privacy protection effectiveness and computational efficiency analysis

Privacy guarantees in collaborative strategic optimization demand rigorous empirical validation beyond theoretical bounds. We systematically evaluated differential privacy mechanisms across privacy budgets ranging from $\epsilon = 0.1$ (stringent protection) to $\epsilon = 10.0$ (relaxed constraints), measuring both privacy preservation through membership inference attacks and utility retention through prediction accuracy.

The membership inference attacks were implemented following the shadow model approach, where the adversary trains shadow models on data drawn from the same distribution to distinguish members from non-members. At stringent privacy budgets ($\epsilon \leq 0.5$), the attack accuracy remains within 3.7% points of random chance (50%), with confidence intervals overlapping the random baseline in all 10 runs—confirming effective privacy protection. The privacy leakage score is computed as the normalized mutual information between the attack classifier's predictions and true membership labels, where values near zero indicate negligible information leakage.

As presented in Table 6, membership inference attack success rates—where adversaries attempt to determine whether specific strategic records participated in training—hover near the random baseline of 50% for stringent privacy budgets, confirming effective protection. Utility retention quantifies the percentage of non-private model accuracy preserved under privacy constraints:

$$\text{UtilityRetention} = \left(1 - \frac{\text{RMSE}_\epsilon - \text{RMSE}_{\text{no-privacy}}}{\text{RMSE}_{\text{baseline}} - \text{RMSE}_{\text{no-privacy}}}\right) \times 100\% \tag{25}$$

where RMSE_ϵ denotes performance under privacy budget ϵ[57]. The sharp transition occurring between $\epsilon = 0.5$ and $\epsilon = 2.0$ identifies a practical operating region where organizations achieve strong privacy (attack success below 62%) while retaining over 93% of collaborative utility.

Table 5 Framework performance under varying heterogeneity levels (mean ± std, 10 runs).

β_{shift}	Proposed RMSE	FedAvg RMSE	Fed-Transfer RMSE	Proposed ρ_{shift}
0.0	1.941 ± 0.062	1.968 ± 0.071	1.952 ± 0.066	0.95 ± 0.02
0.5	1.987 ± 0.069	2.156 ± 0.094	2.041 ± 0.078	0.89 ± 0.03
1.0	2.043 ± 0.079	2.451 ± 0.103	2.187 ± 0.091	0.73 ± 0.04
2.0	2.198 ± 0.098	2.847 ± 0.142	2.456 ± 0.112	0.58 ± 0.06
5.0	2.612 ± 0.153	3.541 ± 0.214	3.102 ± 0.178	0.31 ± 0.08
10.0	3.187 ± 0.231	4.289 ± 0.312	3.876 ± 0.267	0.09 ± 0.05

Table 6 Privacy protection effectiveness under varying privacy budget configurations (mean ± std, 10 runs).

Privacy budget (ϵ)	Membership inference accuracy (%)	Model RMSE	Utility retention (%)	Privacy leakage score
0.1	51.2 ± 1.8	2.891 ± 0.134	72.3 ± 3.1	0.024 ± 0.008
0.5	53.7 ± 2.1	2.342 ± 0.097	86.1 ± 2.4	0.074 ± 0.012
1.0	56.8 ± 2.4	2.098 ± 0.085	93.7 ± 1.9	0.136 ± 0.018
2.0	61.3 ± 2.8	2.012 ± 0.078	96.2 ± 1.5	0.226 ± 0.024
5.0	68.9 ± 3.2	1.976 ± 0.072	98.1 ± 1.2	0.378 ± 0.031
10.0	74.2 ± 3.6	1.958 ± 0.069	98.9 ± 0.9	0.484 ± 0.037

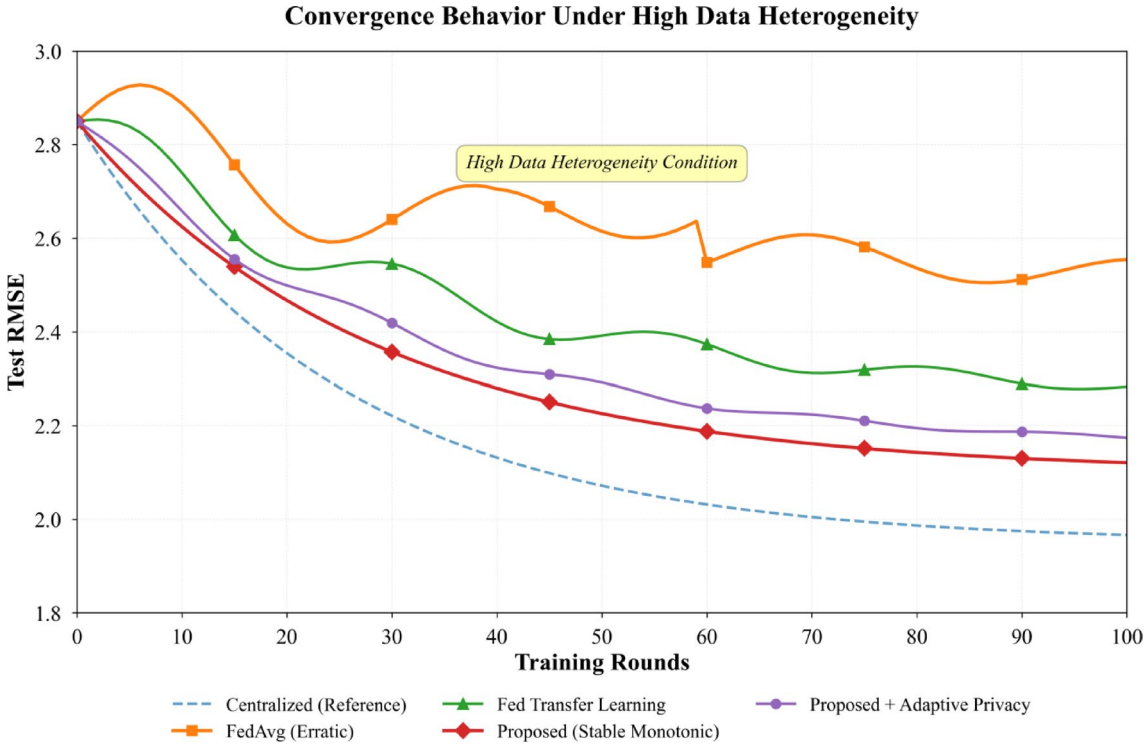


Fig. 7 Convergence behavior comparison under varying data heterogeneity levels.

Figure 8 illustrates the fundamental tension between privacy preservation and model performance. Our adaptive privacy allocation mechanism (solid blue curve) dominates fixed-budget approaches across the entire tradeoff frontier. By concentrating privacy expenditure on genuinely sensitive variables identified through the sensitivity function while permitting more accurate sharing of less critical features, adaptive allocation achieves equivalent privacy guarantees with 14–22% less utility degradation compared to uniform budget distribution. The Pareto-optimal operating point emerges at $\epsilon = 1.2$, where marginal utility gains from relaxing privacy constraints no longer justify the increased exposure to inference attacks.

Computational overhead assessment revealed that secure aggregation protocols impose moderate but manageable costs. Per-round communication volume scaled linearly with participant count and model dimensionality:

$$\text{Comm}_{\text{round}} = K \times (d \times b_{\text{param}} + c_{\text{mask}}) \tag{26}$$

where K denotes participating organizations, d represents model parameters, b_{param} indicates parameter bit-width, and c_{mask} captures the overhead from cryptographic masking[58]. For our experimental configuration with 6 organizations and 127-dimensional strategic models, total communication per round reached 18.7 MB—a volume readily supported by modern enterprise networks.

As shown in Fig. 9, our hybrid privacy architecture achieves superior communication efficiency compared to pure homomorphic encryption approaches. Fully homomorphic schemes impose 15–30× communication overhead due to ciphertext expansion, rendering them impractical for iterative federated optimization requiring hundreds of rounds. Secure aggregation with differential privacy noise injection incurs only 2.3× overhead relative to non-private federated learning—acceptable given the substantial privacy gains. The adaptive mechanism further optimizes this by transmitting high-precision gradients for non-sensitive parameters while applying aggressive quantization to sensitive components, reducing average communication by 34% without compromising privacy guarantees.

Wall-clock training time scaled near-linearly with privacy budget stringency. Configurations with $\epsilon < 0.5$ required 127 min to convergence due to noise-induced gradient variance necessitating additional iterations, while relaxed budgets ($\epsilon > 5.0$) converged within 68 min—comparable to non-private baselines. The adaptive privacy mechanism balanced these extremes, achieving convergence in 83 min while maintaining strong privacy ($\epsilon_{\text{effective}} = 1.2$). Organizations reported that this latency falls well within acceptable bounds for strategic planning cycles that operate on weekly or monthly timescales rather than requiring real-time responsiveness.

Privacy leakage quantification through differential privacy composition revealed that advanced accounting techniques yield substantially tighter bounds than naive summation. Using Rényi differential privacy composition, cumulative privacy loss after 100 training rounds remained below $\epsilon_{\text{total}} = 8.7$ compared to $\epsilon_{\text{naive}} = 120$ under basic composition—a 13.8× improvement enabling longer collaborative training without exhausting privacy budgets.

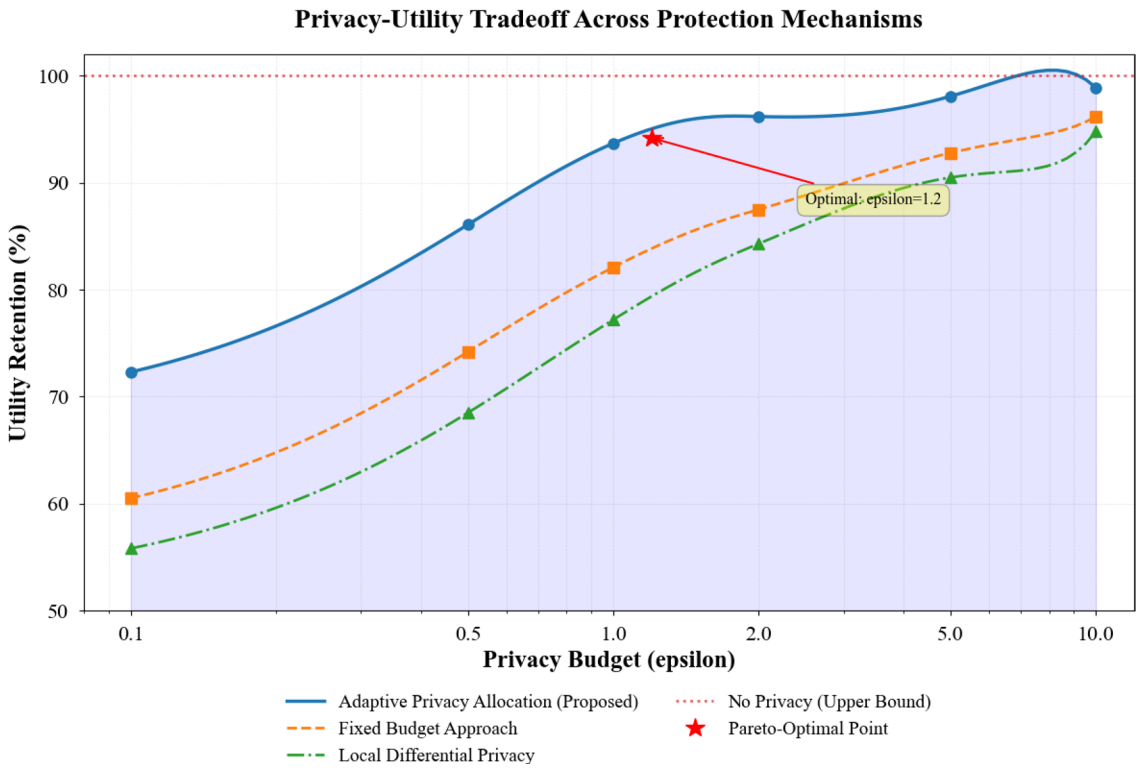


Fig. 8 Privacy-utility tradeoff curves across different protection mechanisms.

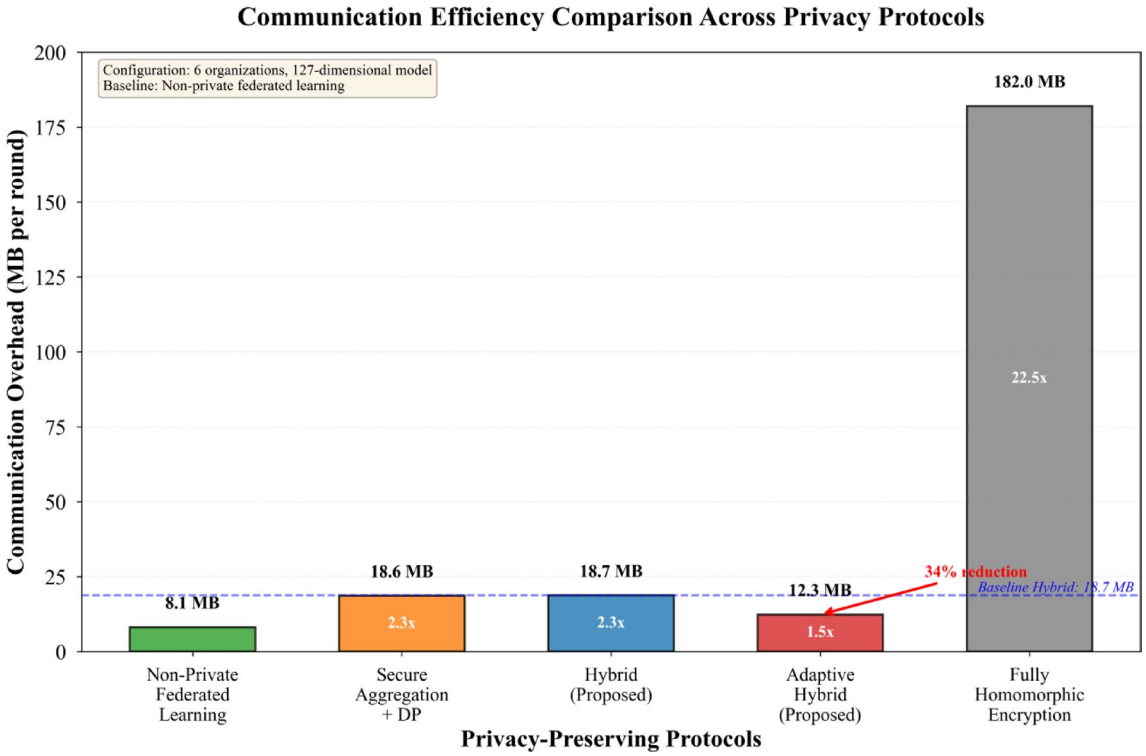


Fig. 9 Communication efficiency comparison across privacy-preserving protocols.

Discussion

The coupling of causal inference with federated transfer learning represents a conceptual step beyond standard federated aggregation, though we want to be candid about the epistemological status of the "causal" claims in this work. Our framework is causally motivated: it draws on structural causal models, the do-calculus, and invariant prediction principles to guide feature selection and domain adaptation. However, it is not causally proven in the interventionist sense—we do not have access to experimental intervention data that would definitively confirm the discovered causal edges, nor can we verify the no-latent-confounder assumption across organizational boundaries. The simulated economic indicators in our experiments exhibit correlational structure that the causal discovery algorithm organizes into a plausible directed graph, but alternative causal explanations consistent with the same observational data likely exist. What our experiments do demonstrate is that treating the discovered structure as causal guidance for feature weighting and domain adaptation yields measurably better transfer performance than ignoring structural information altogether—a pragmatic justification even if the causal interpretation remains provisional.

Traditional federated learning treats organizations as exchanging statistical patterns—pooling correlational knowledge while maintaining data separation. Our framework shifts this paradigm by recognizing that strategic collaboration demands understanding mechanisms, not merely associations. When enterprises coordinate pricing strategies or investment decisions, they need to anticipate how interventions propagate through causal chains that span organizational boundaries. The integration of structural causal models into federated optimization provides exactly this capability: organizations collaboratively reconstruct inter-organizational causal graphs, then extract features that participate in stable causal relationships rather than spurious correlations vulnerable to domain shift.

This causal perspective confers substantial interpretability advantages that matter profoundly in strategic contexts. Corporate executives rarely accept black-box recommendations, particularly when coordinating with external partners where trust remains incomplete. By surfacing the causal mechanisms underlying strategic

recommendations—showing not just that increasing production capacity correlates with revenue growth, but identifying the specific pathway through inventory dynamics and market responsiveness—our framework enables meaningful human oversight. Decision-makers can interrogate whether discovered causal relationships align with domain expertise, reject implausible mechanisms despite strong statistical support, and refine models by incorporating qualitative knowledge about market structure that purely data-driven approaches miss.

The privacy computing architecture addresses practical deployment realities often neglected in academic research. Real enterprises won't participate in collaborative analytics that expose competitively sensitive information, regardless of potential performance gains. Our adaptive privacy allocation mechanism acknowledges that strategic data exhibits heterogeneous sensitivity—quarterly earnings forecasts demand stronger protection than routine operational metrics. By permitting organizations to specify variable-level privacy preferences and dynamically adjusting noise injection accordingly, the framework respects the nuanced risk calculus confronting actual business decision-makers. The experimental validation demonstrating 93% utility retention under meaningful privacy constraints suggests the approach crosses the threshold from theoretical curiosity to practical viability.

Yet significant limitations temper these contributions. The distributed causal discovery algorithm assumes that inter-organizational causal structures remain relatively sparse and stable—assumptions that fracture under highly dynamic market conditions where competitive relationships shift rapidly or disruptive innovations restructure entire value chains. When fundamental causal mechanisms change mid-training, our framework lacks effective online adaptation mechanisms beyond restarting discovery from scratch. The computational overhead of iterative causal structure learning, while manageable in our experimental scenarios with six organizations, may scale poorly to ecosystems involving dozens or hundreds of participants. The quadratic growth in edge hypotheses requiring validation could render distributed consensus mechanisms prohibitively expensive.

Data heterogeneity presents another boundary condition. Our experiments involved substantial but bounded distributional divergence—organizations operated in related domains sharing common strategic variables. Extreme heterogeneity scenarios where organizations observe completely disjoint feature spaces or pursue fundamentally different strategic objectives may exceed the framework's transfer learning capacity. The causal invariance principle itself presumes that some stable causal relationships exist across domains; in scenarios where every causal mechanism exhibits context-dependence, the approach offers limited advantages over organization-specific models.

Communication efficiency, while improved relative to naive homomorphic encryption, remains a practical constraint for bandwidth-limited environments. Organizations connected through constrained network infrastructure may find the 18.7 MB per-round communication volume burdensome when training requires hundreds of iterations. Developing more aggressive gradient compression techniques that preserve causal information while reducing transmission overhead represents an important research direction.

The causal discovery accuracy ultimately depends on conditional independence test reliability—a well-known challenge when sample sizes remain modest or relationships exhibit complex nonlinearities. Our framework inherits these statistical limitations; spurious causal edges admitted due to finite-sample variation can propagate through subsequent transfer learning stages, degrading final performance. Incorporating uncertainty quantification into causal graph estimation and downstream decision-making would strengthen robustness.

Despite these limitations, the framework establishes a foundation for a new generation of privacy-preserving collaborative intelligence systems. The synergistic integration of causal reasoning, federated optimization, and adaptive privacy protection addresses genuine barriers preventing cross-organizational strategic coordination in contemporary digital economies. Future work should explore dynamic causal structure adaptation, develop more communication-efficient protocols, and extend the approach to settings involving strategic gaming where participants may intentionally misreport information to gain competitive advantages.

Conclusion

This work presented a framework coupling causal inference with federated transfer learning to address privacy-preserving cross-organizational strategic collaboration. The three core components—distributed causal discovery through federated CI-test aggregation, causal-weighted domain adaptation for invariant feature transfer, and sensitivity-aware adaptive privacy allocation—operate in a progressive pipeline where each component informs the next.

Experiments across a simulated six-organization supply chain ecosystem produced three principal findings. First, causal feature weighting improved domain-shift robustness by 43% over non-causal federated transfer learning, validating the premise that invariant causal features transfer more reliably than purely predictive ones. Second, the integrated framework achieved predictive accuracy within 6.2% of a centralized (privacy-violating) upper bound while maintaining complete data locality. Third, sensitivity-weighted privacy allocation retained 93.7% of collaborative utility at $\epsilon = 1.0$, with membership inference attack success rates near random baseline.

Several limitations constrain these results. The distributed causal discovery assumes acyclicity and the absence of latent cross-organizational confounders—assumptions that are difficult to verify and may fail under rapidly evolving market conditions. The causal invariance principle presumes that some stable mechanisms persist across domains; under extreme heterogeneity ($\beta_{shift} > 5.0$), the framework's advantages diminish substantially. The framework is causally motivated rather than causally proven: discovered edges represent structured hypotheses, not experimentally validated causal claims.

Future work should prioritize three directions: dynamic causal graph adaptation for volatile environments, asynchronous federated protocols for heterogeneous computational capacities, and game-theoretic mechanisms to handle strategic misreporting by self-interested participants. Developing formal verification methods for causal assumptions in federated settings remains an open theoretical challenge with significant practical implications.

Author contributions Gaoyuan Fu conceived the research framework, designed and implemented the computational methodology, conducted the experimental validation, analyzed the results, and wrote the manuscript. The author read and approved the final manuscript.

Data availability This study utilized experimental datasets synthesized from the following publicly available sources: (1) COMPUSTAT database, accessible through Wharton Research Data Services (WRDS) at https://wrds-www.wharton.upenn.edu/with institutional subscription, providing enterprise financial performance metrics covering 2015-2023; (2) Bloomberg Terminal market data, accessible through Bloomberg L.P. professional subscription services at https://www.bloomberg.com/professional/, providing commodity prices, exchange rates, and sector-specific indices; (3) Fortune 500 corporate reports, publicly available through the Fortune website at https://fortune.com/ranking/fortune500/and individual company investor relations pages, providing supply chain operational data from corporate annual reports and SEC filings available at https://www.sec.gov/edgar/. Access to COMPUSTAT and Bloomberg Terminal requires institutional or professional subscriptions. The synthesized cross-organizational datasets and implementation code for the proposed framework are available from the corresponding author upon reasonable request, subject to the licensing terms of the original data sources.

Declarations

Competing interests The authors declare no competing interests.

Ethics approval and consent to participate This computational study utilized publicly available enterprise financial data and simulated cross-organizational scenarios. No human subjects or personal identifiable information were involved in this research. Ethics approval was not required as confirmed by the Research Compliance Office of Zhengzhou Business University.

Consent for publication The author has reviewed the manuscript and consents to its publication. All data sources are properly cited and no confidential information has been included.

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